RL-ISegNet: Refining Instance Segmentation for Remote Sensing Imagery via Iterative Reward Maximization With Limited Samples

Yang Liu, Tong Zhang, Yanru Huang, Peixiao Wang, and Gang Wang

***Abstract*—Instance segmentation of remote sensing imagery (RSI) is vital for applications like geographic information system (GIS) updates and urban planning. Due to RSI's diversity (e.g., scale variations and complex object shapes), instance segmentation models require extensive labeled data, which is costly and labor-intensive. To mitigate this dependence on extensive labeled data, we propose a reinforcement learning (RL)-driven iteratively-refined instance segmentation model for RSI by RL (RL-ISegNet) framework, which uses policy gradient optimization to enhance the learning of hard samples in limited labeled data, thereby improving data utilization efficiency under limited samples. Specifically, our RL-ISegNet defines a sequential optimization process for hard samples, using cumulative rewards from each subtask (region proposal, detection, and segmentation) to provide future feedback guidance for optimizing network parameters. Meanwhile, considering the conflicts between subtasks, we propose a metric to quantify the positive or negative impacts between subtasks as a loss constraint, thereby increasing the sampling probability of trajectories with positive impacts. Additionally, we developed a Fisher information-based method to freeze low uncertainty parameters, reducing the training time. The experimental results show that our RL-ISegNet improves data utilization efficiency by +12.2% and achieves improvements of +7.1% in instance segmentation metrics with 4% of the training dataset.**

***Index Terms*—Instance segmentation, limited samples, reinforcement learning (RL), remote sensing imagery (RSI).**

I. Introduction

INSTANCE segmentation for remote sensing imagery (RSI) is a fundamental yet challenging task in Earth observation. The RSI instance segmentation aims to achieve both instance-level localization and pixel-level classification of interested instances, which is significant for the automatic updating of geographic information system (GIS) data, traffic monitoring, and urban planning [1], [2], [3]. Currently, the primary challenges in RSI instance segmentation stem from the inherent diversity of RSI, such as scale variations, geographical differences, and arbitrary shapes and sizes of instances. To address the diversity challenges of RSI, many existing instance segmentation models need a substantial quantity of labeled data for training. Insufficient labels may hinder instance segmentation methods from effectively learning data patterns and relationships, leading to diminished accuracy [4], [5], [6], [7]. However, sufficient training samples typically are labor-intensive and costly, making it essential to reduce the dependence of instance segmentation models on the need for extensive labeled samples.

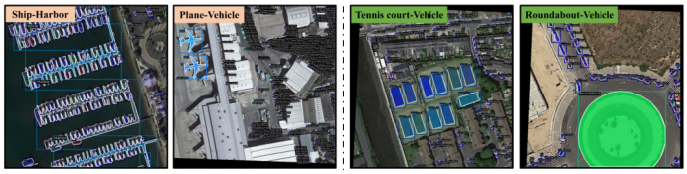


Fig. 1. Challenges of remote sensing target diversity. (a) Challenge of information imbalance. For example, there are significant differences in the number of instances between ships and harbors, and between planes and vehicles. This leads to categories with more instances receiving greater attention from the model during the learning process, resulting in a bias of the model. (b) Challenge of learning difficulty. For example, tennis courts and roundabouts have easily generalizable features (such as clear contours and fixed instance size). In contrast, the size of vehicle instances is smaller, and their features are often disregarded during the convolution process, making it more challenging for the model to learn such instances.

Current research on the mitigation of the dependence on extensive labels in instance segmentation primarily focuses on semi-supervised, weakly supervised, or few-shot methods [4], [5], [8], [9], [10]. These methods leverage the distribution information of unlabeled data [11], the inherent structure of the data [12], or generated pseudo-labels [13] to either enhance data utilization or improve the learning efficiency of labeled data. While these methods alleviate the dependence on labeled data to some degree, significant challenges remain for multiclass instance segmentation in RSI. These challenges mainly stem from the information imbalance among different categories [in Fig. 1(a)] and different learning difficulties among different categories [in Fig. 1(b)]. With limited training samples, these challenges become more pronounced, causing the model to develop a bias toward easily learnable categories while underperforming on hard categories, thereby affecting

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overall segmentation performance [14], [15], [16]. Although some methods [17], [18] incorporate class-balanced loss functions to increase the focus on hard samples. However, hard samples during training may be very limited and insufficient, resulting in the model's inability to fully learn from these hard samples. Therefore, when only limited samples are available, enhancing the learning efficiency for difficult samples is key to diminishing the dependence of instance segmentation models on training samples.

The issue of increasing the utilization efficiency of limited samples has been extensively explored in the field of natural language processing (NLP). For example, the fine-tuning of NLP models uses reinforcement learning (RL) with rewards defined by human feedback to maximize reward signals in limited samples, thereby improving sample learning efficiency [19], [20], [21]. Inspired by the fine-tuning mechanism of NLP models, the computer vision community has also begun to continuously refine object detection or semantic segmentation models via RL, which maximizes rewards on specific refinement functions [22], [23], [24], [25], [26], [27]. These methods achieve the optimization of expected tasks through a continuous trial-and-error process, achieving significant performance improvement. However, since instance segmentation has multiple subtasks (i.e., region proposal network (RPN), detection, and segmentation), optimizing the instance segmentation model via RL is much more challenging than the optimization of a single task. It faces three major challenges.

1) *Reward Function Design:* The reward function for hard samples needs to meticulously capture the complexity and objectives of the instance segmentation subtasks. Due to different objectives of the subtasks, each task has its unique actions, state, and feature spaces, requiring the design of distinct reward functions based on the corresponding requirements for each subtask.
2) *Intertask Conflict Optimization:* The inconsistency in the optimization directions of two or more subtasks can lead to conflicts between them. Attempting to optimize these subtasks simultaneously could result in suboptimal performance across all subtasks on the shared network.
3) *Training Efficiency:* The instance segmentation model uses RL to optimize the network parameters of subtasks through extensive sampling, which significantly increases the training time.

To address the above three challenges, we propose an iteratively refined instance segmentation model for RSI by RL (RL-ISegNet). We define the learning process of hard samples as a sequential optimization process, optimizing current parameters by considering future cumulative rewards, i.e., the accumulated RPN classification scores, the intersection over union (IOU) of detection, and the spatial distance of features. Specifically, our RL-ISegNet method iteratively refines the policy gradient of multiple subtasks to maximize the three cumulative rewards of hard samples. Each subtask uses cumulative rewards to guide the model to find better parameters for hard samples, thereby fully mining information on hard samples to improve the learning efficiency of the model on limited samples and reduce the model's reliance on labels. As shown in the following three points, for challenge 1, we construct three reward functions to address the issues of foreground–background distinction in the RPN subtask, interclass misclassification in the detection subtask, and pixel misclassification in the segmentation subtask. We estimate the gradient of the expected subtask by computing the maximum cumulative rewards that consider the RPN classification scores, detection IOU, and spatial distance of features, thereby optimizing the parameters of the entire model. For challenge 2, we propose a metric as a constraint term that quantifies the positive and negative impacts between subtasks. The constraint is to maximize the sampling probability of positive impacts and minimize the probability of negative impacts for each subtask trajectory, which is a sequence of states, actions, and rewards. The design of the constraint ensures that the optimization of a specific subtask is consistent with other subtasks. For challenge 3, we develop a Fisher information-based method to measure parameter sensitivity and freeze specific parameters to save training time in the complex iterative optimization of multiple subtasks.

In summary, due to the high cost of instance segmentation annotations, the proposed RL-ISegNet aims to improve learning efficiency under limited supervision and reduce reliance on large-scale labeled data. Existing methods that rely on static weighting or mining strategies based on short-term loss signals neglect the long-term contribution of samples in learning, thereby overlooking hard but informative instances. The proposed RL-ISegNet models hard sample learning as a sequential decision process. Using RL to maximize long-term cumulative rewards, the model is guided to prioritize high-reward samples that improve overall performance, thereby reducing the demand for labeled samples. Our main contributions are summarized as follows.

1) To mitigate the problem that short-term loss signals ignore the long-term contribution of hard samples, we propose an iteratively refined instance segmentation framework (RL-ISegNet) for RSI via reward maximization with limited samples. RL-ISegNet models hard sample learning as a sequential optimization process, using cumulative rewards that provide future feedback to guide the model in finding better parameters. This enables the model to fully mine information from hard samples and improve learning efficiency under limited supervision.
2) Inadequate understanding of intertask relationships hampers effective conflict handling and coordinated optimization. To mitigate this problem, we propose a consistency learning method that quantifies intertask relationships to alleviate conflicts between subtasks in instance segmentation. Specifically, we design a metric that explicitly quantifies the positive and negative impacts between subtasks, which serves as a constraint term to facilitate the coordination of each subtask, thereby improving the sampling probability of positive impacts for each subtask trajectory.
3) To reduce the high training overhead of the RL process, it is important to avoid unnecessary parameter updates. Since RL optimizes policies based on expected

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cumulative rewards, it is crucial that parameter freezing methods align with the expected impact on model outputs. Therefore, we propose a Fisher information-based parameter freezing method that links parameter importance to long-term model behavior by quantifying each parameter's expected contribution to the output distribution, reducing the risk of freezing critical parameters.

II. Related Work

A. Instance Segmentation With Limited Samples in RS

In current research on reducing the reliance of vision tasks on large amounts of labeled data, two major strategies have emerged: leveraging pretrained models [28], [29] and designing task-specific networks [30], [31]. In this study, we focus on the latter—developing task-specific designs to reduce label dependence during training. Current research on reducing the dependence on large amounts of labels in instance segmentation [32], [33], [34], [35], [36], [37], [38] mainly focuses on methods such as semi-supervised, weakly supervised, and few-shot learning [4], [5], [8], [9], [10]. Semi-supervised learning enhances the generalization ability of the instance segmentation model by combining a small amount of labeled data with a large amount of unlabeled data [5], [11], [39], [40], [41]. In semi-supervised learning, the quality of pseudo-labels has a significant impact on the outcomes. When pseudo-labels exhibit significant noise, the model performance may deteriorate. Weakly supervised methods guide the learning of an instance segmentation model by utilizing partially labeled data (such as image-level labels, coarse annotations, or imprecise labels) [8], [42], [43], [44], [45]. The methods reduce the demand for high-quality labels by extracting useful information from low-cost coarse labels. Essentially, weak supervision does not reduce the demand for labels, but instead lowers the labeling costs to acquire more annotations for training. Few-shot methods aim to enable effective learning with a minimal amount of training samples [46], [47], [48]. Chen et al. [49] proposed an instance segmentation network based on the segment anything model (SAM) [50] combined with semantic category information. Few-shot instance segmentation learning is relatively concentrated in the visual domain [46], [47], [48], [51] and is relatively scarce in the remote sensing field. Currently, compared to the other two methods, the segmentation accuracy of the few-shot approach is relatively lower.

The above three methods essentially improve data utilization efficiency by mining data from different perspectives. In this article, we define the information mining process of hard samples as a sequential optimization process, optimizing current parameters by considering future cumulative rewards (long-term returns). This approach maximizes the learning of hard samples, reducing the model's dependence on large amounts of labeled data.

B. Vision Tasks With Limited Samples

Instance segmentation inherently combines classification [52], [53], detection [54], [55], and segmentation tasks [56], [57]. Here, we briefly review recent advancements in these three areas under limited supervision. For example, to address the issue of ignoring hard negatives in pseudo-label fine-tuning and to reduce training cost, Huang et al. [52] proposed a hard negative mining method based on a multiple instance learning ranking loss in image classification. To address the issue of background misclassification caused by incomplete annotations in object detection, Yan et al. [54] proposed an efficient training method that can be integrated into both meta-learning and transfer learning frameworks. Hong et al. [56] incorporated the dice loss, and Xu et al. [57] introduced interclass and intraclass balance losses, as well as a unified balance loss, to improve generalization performance. Unlike the above methods that rely on short-term loss signals, we treat sample learning as a sequential decision-making process and employ RL to maximize the long-term cumulative rewards across multiple subtasks (RPN, detection, and segmentation), thereby enhancing the model's ability to learn from samples.

C. RL in Visual Refinement Tasks

RL can guide decision-making in dynamic and uncertain environments through reward signals, refining decisions through trial-and-error processes, thus enhancing model accuracy. Currently, in the field of vision, methods based on RL to optimize object detection, semantic segmentation, and so on, are gradually increasing [22], [23], [24], [25], [26], [27], [58]. For instance, Lee et al. [59] applied reward weighting to the loss function coefficients to align text-to-image models, and Black et al. [60] treated the denoising process of diffusion models as a Markov decision process (MDP) to fine-tune diffusion models using a reward model. In segmentation, Casanova et al. [22] used an RL [deep Q-network (DQN)] approach to select active learning strategies, iteratively optimizing its sampling strategy. Ding et al. [23] transformed the image segmentation process into a series of decision problems, where the prior knowledge (segmentation results) from the previous frame is used to generate the prediction mask for the current frame. Liu et al. [24] proposed a dynamic scale perception framework called GeoAgent, which adaptively captures appropriate contextual information beyond image patches based on different geographic objects. Pinto et al. [25] used RL to align models with task rewards, demonstrating remarkable performance across multiple computer vision tasks such as object detection, panoptic segmentation, and image description. However, in the field of instance segmentation, the integration of RL has been scarcely explored [61]. Compared to single tasks, instance segmentation presents greater challenges in designing reward functions and preventing task optimization conflicts due to its multiple subtasks.

III. Methodology

A. Motivation

In this article, the objective is to find a specific parameter $\theta : \mathcal{X} \rightarrow \mathcal{Y}$ from the distribution space $\mathcal{D}$ under limited sample conditions (i.e., reducing the number of samples N in the dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, where x represents the input data and y represents the corresponding labels), to maximize the

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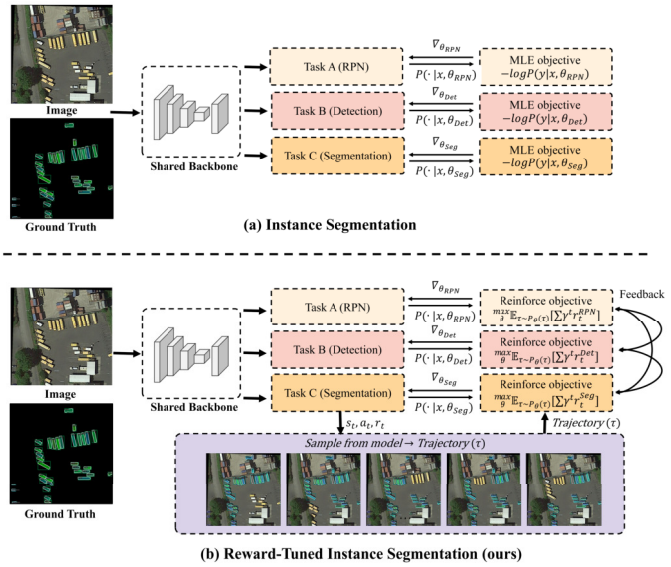


Fig. 2. Comparison of model optimization mechanisms. (a) Instance segmentation model based on maximum likelihood, where the negative log-likelihood functions of the conditional probabilities for each subtask, $P(\cdot \mid x, \theta_{\text{RPN}})$, $P(\cdot \mid x, \theta_{\text{Det}})$, and $P(\cdot \mid x, \theta_{\text{Seg}})$, are used as the objective functions. These objective functions are, then, used to compute the parameter gradients $\nabla_{\theta_{\text{RPN}}}$, $\nabla_{\theta_{\text{Det}}}$, and $\nabla_{\theta_{\text{Seg}}}$ to update the instance segmentation parameters. (b) Proposed RL-ISegNet method. The RL-ISegNet constructs task trajectories τ based on the sampled sequence of states s_t and a_t, and immediate rewards r_t. The cumulative rewards for each subtask G_t^{RPN}, G_t^{Det}, and G_t^{Seg} are computed according to the task trajectories τ. By maximizing these cumulative rewards, the parameter gradients $\nabla_{\theta_{\text{RPN}}}$, $\nabla_{\theta_{\text{Det}}}$, and $\nabla_{\theta_{\text{Seg}}}$ are computed to update the parameters.

performance of the instance segmentation model. However, as N decreases, due to the information imbalance and learning difficulties (as shown in Fig. 1), the model tends to favor the easier categories and perform poorly on the harder categories, which limits the overall segmentation performance. Therefore, we focus on the hard samples within the limited dataset, enhancing the model's information extraction from them to improve its performance.

To enhance the model's learning capability of hard samples in limited training data, we propose a novel RL-driven instance segmentation model [as shown in Fig. 2(b)] through a trial-and-error process that provides future feedback to guide gradient updates for current parameter optimization. Different from the instance segmentation model based on maximum likelihood [as shown in Fig. 2(a)], the proposed model solves for the optimal segmentation parameters θ by maximizing the cumulative rewards of each subtask (RPN, detection, and segmentation), thereby maximizing the overall performance of the instance segmentation model.

B. Notation and Definitions

In this study, we define the corresponding symbols, including the basic elements of policy π_θ, state s_t, action a_t, reward r_t, and task trajectory τ.

Policy (π_θ): The policy function maps a state to an action, denoted as $\pi_\theta(a_t \mid s_t)$. In this article, we aim to optimize the mapping process of the three subtasks from $\mathcal{X} \rightarrow \mathcal{Y}$. To achieve this, we treat the network parameters of the three subtasks as RL policies, represented as $\pi_{\theta_{\text{RPN}}}$, $\pi_{\theta_{\text{Det}}}$, and $\pi_{\theta_{\text{Seg}}}$.

State (s_t): The state is the information input into the policy function at a given step. In this article, we use the multiscale feature $\mathcal{F}$ that is output by the shared backbone at step t as the state $s_t \in \{s_t^{\text{RPN}}, s_t^{\text{Det}}, s_t^{\text{Seg}}\}$.

Action (a_t): The action is the operation that is taken in a given state. Since the objectives of the three subtasks differ, each corresponds to a different action $a_t \in \{a_t^{\text{RPN}}, a_t^{\text{Det}}, a_t^{\text{Seg}}\}$: RPN classifies anchor points into positive or negative categories, detection classifies detection boxes into multiple categories, and mask classifies pixels.

Reward (r_t): The r_t represents the reward obtained by mapping s_t to a_t, providing feedback signals to the model and serving as a key component for parameter optimization. r_t^{RPN}, r_t^{Det}, and r_t^{Seg} denote the immediate rewards for RPN, detection, and segmentation, respectively.

Task Trajectory (τ): The trajectory is a sequence of states, actions, and rewards, i.e., $\tau = \{s_0, a_0, r_0, \ldots, s_{T-1}, r_{T-1}, s_T\}$. In this article, task trajectories are sampled through the policy π_θ of the subtasks, including the trajectories τ^{RPN}, τ^{Det}, and τ^{Seg}. These trajectories are, then, stored in a trajectory buffer [in Fig. 3(f)]. Once the sampling of task trajectories is complete, the corresponding parameters are updated using the task trajectories collected from different subtasks.

Definitions: We consider the three subtask networks (RPN, detection, and segmentation) as the policy networks $\pi_{\theta_{\text{RPN}}}$, $\pi_{\theta_{\text{Det}}}$, and $\pi_{\theta_{\text{Seg}}}$ using the terms of RL. These policy networks construct each subtask trajectory $\tau = \{s_0, a_0, r_0, \ldots, s_{T-1}, r_{T-1}, s_T\}$ based on the sampled sequence of states $s_t \in \{s_t^{\text{RPN}}, s_t^{\text{Det}}, s_t^{\text{Seg}}\}$, actions $a_t \in \{a_t^{\text{RPN}}, a_t^{\text{Det}}, a_t^{\text{Seg}}\}$, and immediate rewards $r_t \in \{r_t^{\text{RPN}}, r_t^{\text{Det}}, r_t^{\text{Seg}}\}$. The cumulative rewards for each subtask G_t^{RPN}, G_t^{Det}, and G_t^{Seg} are calculated according to the task trajectories τ^{RPN}, τ^{Det}, and τ^{Seg}. By maximizing the expected cumulative reward $J(\pi_\theta)$ for the policy network $\pi_\theta \in \{\pi_{\theta_{\text{RPN}}}, \pi_{\theta_{\text{Det}}}, \pi_{\theta_{\text{Seg}}}\}$, we solve for the optimal parameters θ_{RPN}, θ_{Det}, and θ_{Seg} for each subtask, as shown in the following equation:

$$J(\pi_\theta) = \max_\theta \mathbb{E}_{\tau \sim P_\theta(\tau)}[G_t] = \max_\theta \mathbb{E}_{\tau \sim P_\theta(\tau)}\left[\sum_{t=0}^{T} \gamma^t r_t\right] \tag{1}$$

where $J(\pi_\theta)$ is the objective function for the subtasks, representing the expected cumulative reward of the policy networks. $\pi_\theta \in \{\pi_{\theta_{\text{RPN}}}, \pi_{\theta_{\text{Det}}}, \pi_{\theta_{\text{Seg}}}\}$ is the policy network for each subtask. $G_t \in \{G_t^{\text{RPN}}, G_t^{\text{Det}}, G_t^{\text{Seg}}\}$ represents the cumulative reward at step t for the corresponding task trajectory $\tau \in \{\tau^{\text{RPN}}, \tau^{\text{Det}}, \tau^{\text{Seg}}\}$. $r_t \in \{r_t^{\text{RPN}}, r_t^{\text{Det}}, r_t^{\text{Seg}}\}$ denotes the immediate reward at step t, γ is the discount factor, and T is the time step for the task trajectory.

Then, we obtain the policy gradient by differentiating the objective function $J(\pi_\theta)$ for the corresponding subtask. The parameters of each subtask are updated using the formula $\theta \leftarrow \theta + \alpha \nabla_\theta J(\pi_\theta)$, thereby optimizing the subtask parameters. The equation for $\nabla_\theta J(\pi_\theta)$ is shown as follows:

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim P_\theta(\tau)}\left[\sum_{t=0}^{T}\left(\sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}\right) \nabla_\theta \log \pi_\theta(a_t \mid s_t)\right] \tag{2}$$

where $\nabla_\theta J(\pi_\theta)$ represents the gradient of the subtask policy parameters θ. $\sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}$ denotes the cumulative reward from

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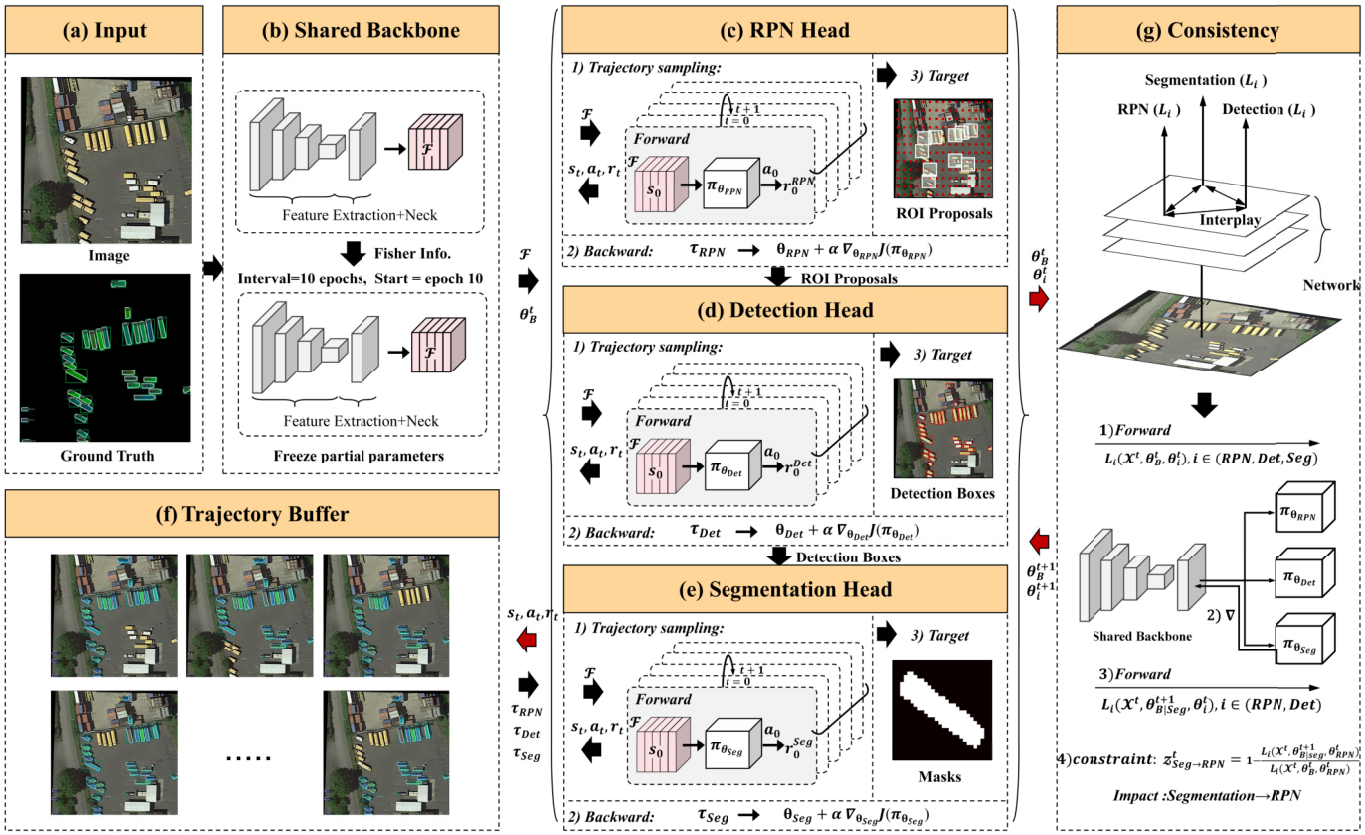


Fig. 3. Framework of RL-ISegNet. (a) Data input. (b) Shared backbone (parameters are denoted as θ_B^t), including the feature extraction network (ConvNeXt-V2 backbone [35]) and the neck network (FPN [62]). Based on the shared backbone, we freeze certain parameters using Fisher information [63], ultimately outputting multiscale features $\mathcal{F}$. To avoid inaccurate estimation in the early training stage and overly frequent adjustments to frozen parameters, we start the freezing process at the tenth epoch and repeat it at intervals of ten epochs. (c) Learning process of the RPN head. In (c-1), the RPN head takes the $\mathcal{F}$ as s_0 and uses the output of the $\pi_{\theta_{\text{RPN}}}$ as the a_0. Based on the a_0, the corresponding reward r_0^{RPN} is calculated, resulting in a task trajectory τ_{RPN} that includes the state, action, and reward. (c-2) Backpropagation begins after the task trajectory τ_{RPN} is collected. (c-3) Generate ROI proposals. (d) and (e) Similar to (c), but the purposes are different: the detection head generates detection boxes, while the segmentation head generates masks. The structural relationship among modules (c)–(e) is shown in Fig. 5. (f) Buffer pool for τ_{RPN}, τ_{Det}, and τ_{Seg}, used to store the trajectories of the three subtasks. (g) Intertask consistency constraint. For example, we compute the impact of the segmentation head on the RPN head. In (g-1), the losses $L_i(\mathcal{X}^t, \theta_B^t, \theta_i^t)$ for the three subtasks are calculated. In (g-2), the loss from the segmentation head is used to update the shared backbone parameters. In (g-3), the updated shared backbone parameters are used to calculate the RPN loss. Then, (g-4) is used to calculate $\mathcal{Z}^t_{Seg\to\text{RPN}}$, which represents the influence of the segmentation head on the RPN head. We use this $\mathcal{Z}^t_{Seg\to\text{RPN}}$ as a constraint to scale the loss $L_i(\mathcal{X}^t, \theta_B^t, \theta_i^t)$. Finally, we update θ_B^t and θ_i^t (where $i \in \{\text{RPN}, \text{Det}, \text{Seg}\}$) based on the constraint $\mathcal{Z}^t_{i\to j}$, obtaining θ_B^{t+1} and θ_i^{t+1}. The red arrows indicate operations performed during the trajectory sampling phase (see Section IV-D for more details).

the current time step to the last time step, where γ is the discount factor. $\sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}$ accumulates future immediate rewards, providing future feedback to guide gradient updates for current parameter optimization. When $\sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}$ is larger, it indicates greater accumulated future rewards, resulting in larger gradient updates, and vice versa. $\nabla_\theta \log \pi_\theta(a_t \mid s_t)$ is the logarithmic gradient of the subtask policy. It represents the gradient of the log probability of selecting the action $a_t \in \{a_t^{\text{RPN}}, a_t^{\text{Det}}, a_t^{\text{Seg}}\}$ under the given state $s_t \in \{s_t^{\text{RPN}}, s_t^{\text{Det}}, s_t^{\text{Seg}}\}$, according to the current subtask policy $\pi_\theta \in \{\pi_{\theta_{\text{RPN}}}, \pi_{\theta_{\text{Det}}}, \pi_{\theta_{\text{Seg}}}\}$.

C. Framework Overview

In this article, we propose the RL-ISegNet framework for instance segmentation for RSI, which is driven by RL to leverage the information from hard samples within limited training data through a trial-and-error process. As shown in Fig. 3, the framework consists of modules 3(a)–(g), with the function of each module explained in the figure caption. This study is centered around RL and focuses on the following three tasks.

1) *Proposing a Parameter Freezing Strategy Based on Fisher Information [see Fig. 3(b)]:* In RL, the trial-and-error process requires multiple steps of experimentation and feedback to find the optimal strategy, which is time-consuming. To reduce the time consumption during the trial-and-error process, we freeze a subset of parameters based on Fisher information and perform RL on top of the frozen structure.
2) *Reward Optimization for the Three Subtasks in Fig. 3(c)–(e):* The reward optimization for the three subtasks guides the model's training based on reward signals. By constructing reward functions for hard samples, the model parameters are continuously adjusted to reduce the error in regions that are challenging to segment, progressively improving the overall model

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performance while maximizing the use of existing labeled data.

3) *Enforcing Intertask Consistency Constraints in Fig. 3(g):* In multitask learning, competition for priority among different subtasks can lead to conflicts between tasks. The intertask consistency constraints aim to resolve intertask conflicts caused by inconsistent optimization objectives across multiple subtasks.

D. Parameter Freezing Based on Fisher Information

In this study, the increased time consumption primarily results from multiple backpropagations during task trajectory sampling. Since the policy network must sample each input multiple times, this leads to significant computational overhead. In the overall framework, the number of parameters in the backbone far exceeds that of the three task heads, resulting in the backpropagation time of the backbone being significantly higher than that of the three task heads. Additionally, since the task heads are responsible for executing specific tasks and require more parameters to summarize features, we only freeze the parameters of the backbone in this study. Therefore, we propose a method based on Fisher information to measure the sensitivity of the backbone parameters, allowing the adaptive freeze of certain parameters to reduce the training parameter number of the model.

Fisher information [63], [64] is an important concept in statistics used to measure the precision of parameter estimates. It describes the sensitivity of a parameter to changes in the likelihood function under given data. Fisher information quantifies the uncertainty of a parameter in the data generation process. Specific definition: let $p(x \mid \theta)$ be a probability distribution controlled by the backbone parameter θ. The Fisher information matrix $I(\theta_i)$ can be defined as

$$I(\theta_i) = -E\left[\frac{\partial^2 log p(x \mid \theta)}{\partial \theta_i^2}\right] \tag{3}$$

where $logp(x \mid \theta)$ is the logarithm of the likelihood function $p(x \mid \theta)$. $(\partial logp(x \mid \theta)/\partial\theta_i)$ is the first derivative of the log-likelihood function to the parameter θ, representing the sensitivity of the data x to the parameter θ under that parameter. This derivative is known as the score function, which describes the impact of parameter changes on the likelihood. $(\partial^2 logp(x \mid \theta)/\partial\theta_i^2)$ is the second derivative of the log-likelihood function to the parameter θ, indicating the rate of change of the score function to θ. A negative second derivative reflects the curvature of the likelihood function. The greater the curvature, the smaller the uncertainty in parameter estimation.

In this study, we assess the sensitivity of the backbone parameters using the Fisher information matrix to determine which parameters can be frozen during training. If $I(\theta_i)$ is high, it indicates that the parameter has a significant impact on the model's output. Therefore, we set a threshold, and if the Fisher information of a parameter is below this threshold, we consider that parameter to be eligible for freezing. To avoid inaccurate estimation in the early training stage and overly frequent adjustments to frozen parameters, we initiate the freezing process at the tenth epoch and repeat it at intervals of ten epochs. The subsequent RL process is, then, conducted on top of the frozen network structure.

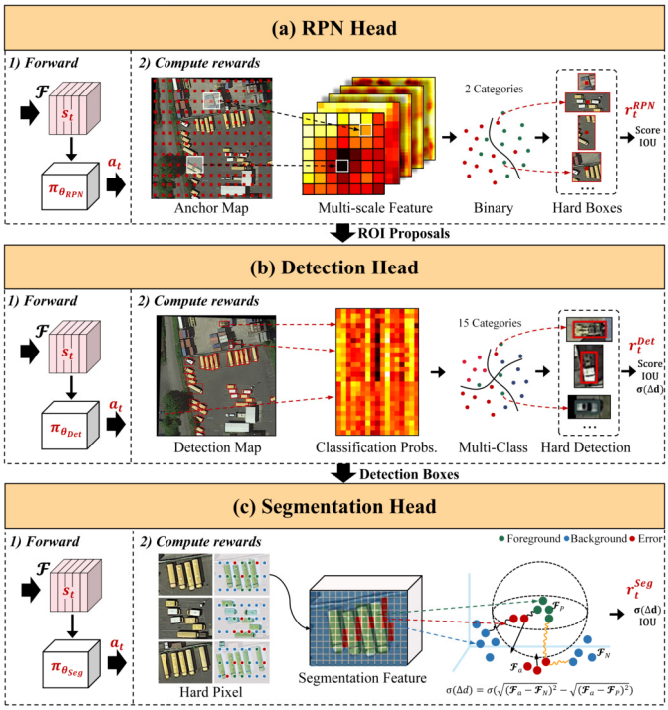


Fig. 4. Reward functions for hard samples in the three subtasks. (a) Anchor points classified incorrectly by the binary classification in the multiscale features obtained through the $\pi_{\theta_{\text{RPN}}}$ policy network as hard samples. The classification scores and IOU of these hard samples serve as the immediate reward r_t^{RPN} for the RPN head at each step. (b) Detection boxes classified incorrectly in the multiclass probability map obtained through the $\pi_{\theta_{\text{Det}}}$ policy network as hard samples. The classification scores, IOU, and $\sigma(\Delta d)$ of these hard samples serve as the immediate reward r_t^{Det} for the detection head at each step. (c) Segmentation head calculates $\sigma(\Delta d)$ and IOU metrics from the foreground features, background features, and erroneous features obtained through the $\pi_{\theta_{\text{Seg}}}$ policy network. These values serve as the immediate reward r_t^{Seg} for the segmentation head at each step. The sphere represents the feature space.

E. Reward Function for Subtasks

In this section, we provide a detailed introduction to the three distinct reward functions in the instance segmentation network (as shown in Fig. 4), which are designed for different subtasks (RPN head, detection head, and segmentation head in Fig. 5).

1) Reward Function for the RPN Head: The RPN head is used to extract a set of region of interest (ROI) proposals from the original image. However, in complex backgrounds or when object edges are not clear, the RPN often struggles to accurately distinguish between foreground and background, leading to missed detections where some foreground objects are not correctly identified, thus affecting downstream detection and segmentation performance. To alleviate this issue, we have constructed a corresponding reward function for hard examples in the RPN [as shown in Fig. 4(a)], aiming to maximize the accuracy of the candidate boxes extracted by the RPN by distinguishing between foreground and background.

To enhance the distinction between the foreground and background, we consider ROI proposals that intersect with

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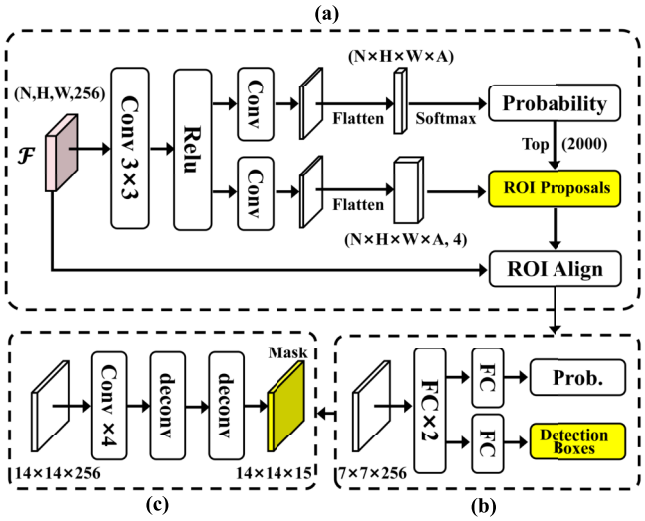


Fig. 5. Network structure diagrams of the three subtasks. (a) RPN network, where A represents the number of anchors, and the top 2000 box coordinates are selected based on probability as ROI proposals. The ROI proposals are processed through ROI align to unify the ROI features to a fixed size. (b) Detection network, which uses fully connected (FC) layers to obtain the class probabilities of the ROI proposals and adjust the box coordinates within the ROI proposals. (c) Segmentation network, which uses four 1×1 convolutions and two 2×2 deconvolutions to generate the mask results. In the diagram, "conv" refers to a 1×1 convolution.

positive samples but are misclassified as hard samples. We calculate their probability scores and IOU as a reward function [as shown in (4)]. Then, we use the cumulative rewards to assign weighted updates to the RPN head, thereby enhancing its focus on hard ROI proposals

$$r_t^{\mathrm{RPN}} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{1+e^{-Z_i}}\right) + \frac{1}{N}\sum_{i=1}^{N}(\mathrm{IOU}_i) \tag{4}$$

where Z_i is the logit value of the ith ROI proposal, calculated from the output layer of the RPN network. It represents the confidence of the model that the box is foreground. The larger the value of Z_i, the more the model believes that proposal i is foreground, conversely. IOU_i represents the intersection over union of the ROI proposal with the ground-truth box. N denotes the total number of proposals.

Then, the parameters θ_{RPN} are updated based on the immediate reward r_t^{RPN} using (2), i.e., $\theta_{\mathrm{RPN}} \leftarrow \theta_{\mathrm{RPN}} + \alpha\nabla_{\theta_{\mathrm{RPN}}}\left(\max_\theta \mathbb{E}_{\tau\sim P_\theta(\tau)}\left[\sum_{t=0}^{T}\gamma^t r_t^{\mathrm{RPN}}\right]\right)$.

2) Reward Function for the Detection Head: In the detection phase of the instance segmentation task, misclassification between different categories is a common issue. This occurs because features of certain categories may be very similar, making it difficult for the model to correctly distinguish between them, thereby affecting the overall detection accuracy. As shown in Fig. 4(b), to alleviate this problem, we have constructed a reward function for hard samples in detection (multiclass misclassified ROI candidate boxes), focusing on maximizing the model's classification accuracy across different categories.

The detection head performs multiclass classification and bounding box regression based on the ROI proposals provided by the RPN head. It faces the challenge of distinguishing multiple categories. Compared to the binary classification problem of the RPN head, the detection head is prone to confusion among similar categories (e.g., small vehicles and large vehicles), primarily because the differences between these two categories are minor, placing them relatively close in feature space and making them difficult to separate. To alleviate this issue with multiclass hard samples, we not only consider the probability scores and IOU from (4), but also introduce a triplet similarity metric [as shown in (6)], which reduces the distance to positive samples and increases the distance to negative samples. The reward function of the $\pi_{\theta_{\mathrm{Det}}}$ policy network is expressed as

$$r_t^{\mathrm{Det}} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{1+e^{-Z_i}}\right) + \frac{1}{N}\sum_{i=1}^{N}(\mathrm{IOU}_i) + \frac{1}{N}\sum_{i=1}^{N}(\sigma(\Delta d_i)) \tag{5}$$

where Z_i is the logit value of the ith detection box, calculated from the output layer of the detection network. It represents the model's confidence that the box belongs to a certain category. IOU_i represents the intersection over union of the detection boxes with the ground-truth box. $\sigma(\Delta d_i)$ is the triplet similarity metric. N denotes the number of detection boxes

$$\sigma(\Delta d_i) = \frac{1}{1+e^{-(s(a_i,p_i)-s(a_i,n_i))}} \tag{6}$$

$$s(a_i, p_i) = \frac{a_i \cdot p_i}{\|a_i\|\,\|p_i\|} \tag{7}$$

$$s(a_i, n_i) = \frac{a_i \cdot n_i}{\|a_i\|\,\|n_i\|} \tag{8}$$

where $\sigma(\Delta d_i)$ is the triplet similarity metric, where $s(a_i, p_i)$ represents the cosine distance between the feature of the misclassified detection boxes and the correct category. a_i denotes the feature vector of the ith misclassified detection box, p_i represents the feature vector of the detection box for the correct category, and n_i denotes the feature vector of the detection box for the incorrect category. When $\sigma(\Delta d_i)$ approaches 1, it indicates that the feature space of the misclassified detection box is getting closer to the correct category.

Then, the parameters θ_{Det} are updated based on the immediate reward r_t^{Det} using (2), i.e., $\theta_{\mathrm{Det}} \leftarrow \theta_{\mathrm{Det}} + \alpha\nabla_{\theta_{\mathrm{Det}}}\left(\max_\theta \mathbb{E}_{\tau\sim P_\theta(\tau)}\left[\sum_{t=0}^{T}\gamma^t r_t^{\mathrm{Det}}\right]\right)$.

3) Reward Function for the Segmentation Head: The segmentation head is used to generate precise pixel-level classification masks for each detected object. However, in practical applications, pixel-level misclassification issues persist, which can lead to inaccuracies in the segmentation masks and ultimately affect the quality of instance segmentation. As shown in Fig. 4(c), to reduce pixel-level misclassifications, we have constructed a reward function for hard objects in the masks (misclassified pixels), aiming to maximize pixel classification accuracy within each detection box.

Specifically, the segmentation head performs pixel classification based on the detection boxes provided by the detection head. During the segmentation process, if the foreground objects closely resemble the background in complex scenes,

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the segmentation head may fail to produce effective segmentations. Similar to the detection head, we also need the triplet similarity metric to reduce the distance to positive samples while increasing the distance to negative samples. However, unlike the detection head, which obtains feature vectors of the same size through ROI align, the distribution of misclassified pixels in segmentation is irregular, making it impossible to directly calculate the triplet similarity metric. To address this, we average the vectors of different categories to obtain average vector representations for misclassified pixels, correct categories, and incorrect categories. The reward function for the segmentation head is shown in (9)

$$r_t^{\text{Seg}} = \frac{1}{N}\sum_{i=1}^{N}(\text{IOU}_i) + \frac{1}{N}\sum_{i=1}^{N}(\sigma(\Delta d_i)) \tag{9}$$

where IOU_i represents the intersection over union of the masks and the ground-truth masks. $\sigma(\Delta d_i)$ is the triplet similarity metric [as shown in (10)]. N represents the number of masks

$$\sigma(\Delta d_i) = \frac{1}{1 + e^{-(s(a_{ij},p_{ij}) - s(a_{ij},n_{ij}))}} \tag{10}$$

$$s\left(a_{ij}, p_{ij}\right) = \frac{a_{ij} \cdot p_{ij}}{\left\|a_{ij}\right\| \left\|p_{ij}\right\|} \tag{11}$$

$$s\left(a_{ij}, n_{ij}\right) = \frac{a_{ij} \cdot n_{ij}}{\left\|a_{ij}\right\| \left\|n_{ij}\right\|} \tag{12}$$

where $\sigma(\Delta d_i)$ is the triplet similarity metric. a_{ij} represents the average vector for the jth misclassified pixel in the ith detection box, p_{ij} denotes the average vector corresponding to the correct category for a_{ij}, and n_{ij} represents the average vector corresponding to the incorrect category for a_{ij}.

Then, the parameters θ_{Seg} are updated based on the immediate reward r_t^{Seg} using (2), i.e., $\theta_{\text{Seg}} \leftarrow \theta_{\text{Seg}} + \alpha\nabla_{\theta_{\text{Seg}}}\left(\max_\theta \mathbb{E}_{\tau\sim P_\theta(\tau)}\left[\sum_{t=0}^{T}\gamma^t r_t^{\text{Seg}}\right]\right)$.

F. Intertask Consistency Constraints

The main reason for the conflicts in multitask learning is the competition for priority between different subtasks, which may lead to a bias toward one subtask when directly summing losses. The rough weight allocation by directly summing losses can overlook the gradient contributions of other tasks, resulting in suboptimal model performance. The challenge with multitask conflicts lies in addressing the negative impacts of joint learning tasks. Ideally, subtasks should be mutually beneficial, promoting each other and maintaining consistency in optimization (maximizing the overall network expectation). Therefore, it is crucial to clarify the interactions between subtasks and guide their optimization toward beneficial directions using positive and negative influences (as shown in Fig. 6). While some methods already leverage uncertainty to determine the weight relationships of loss functions [65], [66], [67], [68], the relationships between subtasks remain unclear. To address this, we propose a metric to quantify the positive and negative influences between subtasks We utilize this quantitative metric to construct constraint terms between subtask trajectories, allowing each subtask trajectory to maximize the sampling probability of positive influences and minimize that of negative influences, thereby guiding the tasks toward beneficial optimization directions (as shown in Fig. 6).

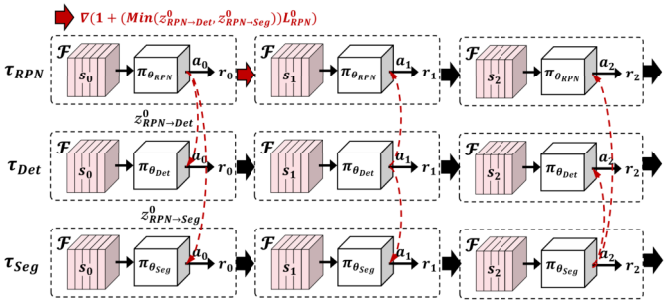


Fig. 6. Consistency constraints between subtask trajectories. $Z^t_{\text{RPN}\to\text{Det}}$ is the quantified influence of the RPN head on the detection head at step $t = 0$. $Z^t_{\text{RPN}\to\text{Seg}}$ is the quantified influence of the RPN head on the segmentation head. For the RPN head, we select the minimum value from $Z^t_{\text{RPN}\to\text{Det}}$ and $Z^t_{\text{RPN}\to\text{Seg}}$ as the constraint term for the loss function L^0_{RPN}. The quantified values for the other subtasks can be similarly derived from the RPN head. $Z^t_{i\to j} > 0$, it indicates that the gradient update from task i reduces the loss of task j, representing a positive influence, otherwise.

Given the subtask set $T = \{RPN, \text{Detection, and Segmentation}\}$, the network backbone parameters θ_B, and the network head parameters θ_i for subtasks in T, we update the influence of subtask i on subtask j by first using the loss of task i [32] to perform a gradient update on the backbone parameters θ_B, expressed as follows:

$$\theta_{B|i}^{t+1} = \theta_B^t - \alpha\nabla_{\theta_B^t} L_i\left(\mathcal{X}^t, \theta_B^t, \theta_i^t\right) \tag{13}$$

where $L_i(\mathcal{X}^t, \theta_B^t, \theta_i^t)$ is the loss corresponding to task $i \in \{\text{RPN, Det, Seg}\}$ under the backbone parameters θ_B, and this loss L_i is used to perform a gradient update on the backbone parameters. $\mathcal{X}^t$ is the action of task i at time t. $\theta_{B|i}^{t+1}$ represents the backbone parameters updated using the loss L_i of task i. We can, then, define the constraint term $\mathcal{Z}_{i\to j}^t$, which represents the positive and negative influences of task i on task j, as shown in the following formula:

$$\mathcal{Z}_{i\to j}^t = 1 - \frac{L_i\left(\mathcal{X}^t, \theta_{B|i}^{t+1}, \theta_j^t\right)}{L_i\left(\mathcal{X}^t, \theta_B^t, \theta_j^t\right)} \tag{14}$$

where $L_i(\mathcal{X}^t, \theta_{B|i}^{t+1}, \theta_j^t)$ is the new loss for task $j \in \{\text{RPN, Det, Seg}\}$ based on the updated backbone parameters from task i, while $L_i(\mathcal{X}^t, \theta_B^t, \theta_j^t)$ is the old loss for task j. $\mathcal{X}^t$ is the action of task j at time t. If $\mathcal{Z}_{i\to j}^t > 0$, it indicates that the gradient update from task i reduces the loss of task j, representing a positive influence, otherwise. We clip $\mathcal{Z}_{i\to j}^t$ to the range of -1 to 1. After calculating $\mathcal{Z}_{i\to j}^t$, we need to restore $\theta_{B|i}^{t+1}$ back to θ_B^t.

Finally, we update the gradient of subtask i based on the quantified influence value on subtask j by selecting the minimum value of $\mathcal{Z}_{i\to j}^t$ (as shown in Fig. 6) as a constraint to adjust the gradient computed for task i, as shown in the following formula:

$$\theta_B^{t+1} = \theta_B^t - \alpha\nabla_{\theta_B^t}\left(\left(1 + Z_{i\to j}^t\right) * L_i\left(\mathcal{X}^t, \theta_B^t, \theta_i^t\right)\right) \tag{15}$$

where $L_i(\mathcal{X}^t, \theta_B^t, \theta_i^t)$ is the loss for subtask i under the backbone parameters θ_B, $Z_{i\to j}^t$ is the constraint term, α is the learning

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rate, θ_B^t is the parameters of the backbone network at time step t, and θ_B^{t+1} is the parameters at time step $t+1$. This multitask consistency constraint process is executed during the task trajectory sampling phase [i.e., the processes in Fig. 3(c-1), (d-1), and (e-1)].

IV. Experimental Settings

A. Experimental Parameter Settings

In this article, the model is trained on an Ubuntu system using a single A100 GPU with a compilation environment including PyTorch 1.13.1, CUDA 11.6, and Python 3.10. The learning strategy for the model is as follows: we use the AdamW optimizer with an initial learning rate set to 0.0001, weight decay of 0.05, and beta parameters of (0.9, 0.999) in AdamW. Each mini-batch consists of four images, and the model is trained for a total of 100 epochs. Since RL in this article involves multiple samplings (with a task trajectory length of 5), the comparative algorithms are trained for 500 epochs to ensure the same number of sampling instances.

B. Evaluation Metrics

We employ standard COCO evaluation metrics for performance assessment [69], which can be specifically categorized based on IOU thresholds into AP, AP_{50}, and AP_{75}. AP refers to the mean average precision calculated over IOU thresholds ranging from 0.5 to 0.95 at intervals of 0.05. AP_{50} and AP_{75} are computed at IOU thresholds of 0.5 and 0.75, respectively. In this article, we evaluate the model's detection and segmentation results using AP_{box}, AP^{50}_{box}, AP^{75}_{box}, AP_{mask}, AP^{50}_{mask}, and AP^{75}_{mask}. Additionally, we report the model's parameter count, floating point operations (FLOPs), and frames per second (FPS) metrics. FLOPs measure computational complexity, while FPS reflects inference speed in frames/s.

C. Dataset

To evaluate the performance of our method under limited labeled data conditions, we designed a series of progressively increasing training set proportions (1%, 4%, 7%, and 10%) to assess the model's performance. Our primary motivation is to maximize accuracy with limited supervision, aiming to approach the performance of full-data (100%) training while substantially reducing labeling costs. Based on the experimental results and considering the performance gap compared to full-data training, we selected 4% and 7% as the primary proportions for in-depth analysis.

In this article, we evaluate the performance of our method on the ISAID dataset [70] (with additional evaluations on two other datasets provided in the Appendix). ISAID is a large-scale remote sensing dataset (e.g., 4000 pixels in width) specifically constructed for instance segmentation, containing numerous small objects and complex backgrounds. It includes 15 object categories. Considering the hardware challenges posed by large-scale images, we train and evaluate baseline methods on 800×800 image patches extracted from the original large-scale images, with a stride of 200. During training, the cropped 800×800 patches are resized to 512×512.

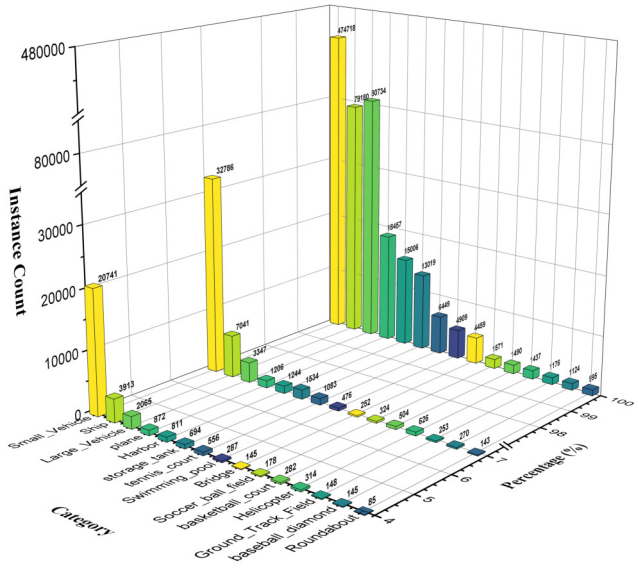


Fig. 7. Number of instances per category at different proportions. This shows the number of instances corresponding to different proportions of the ISAID training set, categorized into 4%, 7%, and 100% training set ratios, which correspond to 750, 1311, and 18 732 images, respectively.

In total, we obtain 18 732 training images (as shown in Fig. 7) and 1698 test images. The training data include both detection and segmentation labels. In this article, supervised methods use both detection and segmentation labels, semi-supervised methods use the same labels with additional unlabeled images, and weakly supervised methods use only detection labels.

D. Implementation Details

In this article, we construct our RL framework RL-ISegNet based on the three-head network structure of [32] for RPN, detection, and segmentation networks, and the backbone from [35]. The entire framework learning process can be divided into the following steps.

1) At intervals of ten epochs, we compute the Fisher information $I(\theta_i)$ for each parameter and then freeze parameters where $I(\theta_i) < 5e^{-7}$. Subsequent training will be based on these frozen parameters. While freezing parameters can reduce training time, it may also lead to some accuracy loss. Therefore, we only compared the time consumption before and after freezing parameters using Fisher information Section VI-G6.
2) In the RL part, we first collect the state–action–reward sequences for each subtask $\pi_{\theta_{\text{RPN}}}$, $\pi_{\theta_{\text{Det}}}$, and $\pi_{\theta_{\text{Seg}}}$ during forward propagation, forming the task trajectory $\{s_0, a_0, r_0, \ldots, s_{T-1}, r_{T-1}, s_T\}$ with $T = 5$, which includes three task trajectories τ^{RPN}, τ^{Det}, and τ^{Seg} [as shown in Fig. 3(f)]. During the sampling process, we need to interact actions with states (i.e., the state generated after the action is executed by policy π changes at step $t+1$. Thus, we use gradient updates to adjust each task's parameters based on their respective losses [32]). Before sampling a task trajectory, we save all network weight parameters, and after sampling, we replace the updated network weight parameters with

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TABLE I
COMPARISON OF INSTANCE SEGMENTATION MODEL ACCURACY (4% ISAID TRAINING SET). BOLD INDICATES THE BEST RESULTS AND UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS IN THE TABLE

Method	Backbone	#Params	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg (%)	FLOPs	FPS
Mask-RCNN	Resnet-101	63.1M	24.7	39.1	26.9	19.7	36.1	18.7	27.5	134G	45.9
Mask2Former	Resnet-101	63.0M	22.7	37.4	23.5	20.3	36.5	19.1	26.6	85.6G	23.5
RTMDet-Ins-x	CSPNeXt	102.7M	<u>37.4</u>	57.4	40.0	28.1	<u>49.9</u>	26.3	39.9	183G	31.4
ConvNeXt-V2	ConvNeXt-V2-B	110.6M	36.1	53.2	<u>40.8</u>	<u>29.7</u>	49.2	<u>30.5</u>	<u>39.9</u>	172G	25.1
PAIS	Resnet-101	56.4M	17.6	34.8	15.5	15.7	31.1	13.9	21.4	69.4G	5.9
GuidedDistillation	Resnet-101	62.9M	18.1	31.6	17.9	21.9	38.9	21.2	24.9	171.3G	13.7
Noisyboundaries	Resnet-101	62.82M	21.2	39.7	20.5	17.8	35.0	15.6	25.0	134.6G	28.3
BoxInst	Resnet-101	53.98M	25.8	42.5	27.0	12.5	28.9	9.5	24.4	115G	24.9
Discobox	Resnet-101	65.06M	17.3	31.5	16.2	15.8	30.5	14.4	21.0	124.7G	29.6
Apro_box2mask	Resnet-101	62.89M	24.3	44.4	23.1	10.9	28.0	7.1	23.0	78.5G	15.9
BoxSnake	Resnet-101	45.9M	30.6	49.3	32.9	20.8	39.1	18.7	31.9	136.5G	19.7
Box2Mask	Resnet-101	62.9M	28.6	48.0	28.3	18.3	37.8	15.8	29.5	78.9G	15.5
Rsprompter	ViT-L	320.2M	26.1	10.3	27.4	23.2	39.9	23.0	25.0	664.5G	6.5
BoxSnake	Swin-L	217.5M	36.9	<u>61.6</u>	38.5	24.0	46.9	21.7	38.3	607.1G	8.6
Box2Mask	Swin-L	215.5M	33.9	54.8	34.4	23.3	44.8	21.4	35.4	234.4G	13.6
Ours	ConvNeXt-V2-B	110.6M	**42.3**	**62.3**	**47.0**	**35.4**	**58.5**	**36.6**	**47.0**	172G	25.1

the saved weights. The multitask consistency constraint ensures consistency across tasks during gradient updates, utilizing the constraint $Z^t_{i\to j}$ [as shown in (14)].

3) We compute the cumulative reward (where the discount factor γ is set by 0.9) at time t based on the immediate rewards r_t^{RPN}, r_t^{Det}, and r_t^{Seg} [as shown in Fig. 3(c), (d), and (e)].
4) Once the task trajectories are collected, we begin backpropagation, updating subtask parameters using the formula $\theta \leftarrow \theta + \alpha\nabla_\theta J(\pi_\theta)$.

V. RESULTS

To evaluate our proposed supervised instance segmentation method, we conducted comparative experiments on the ISAID dataset with four other supervised methods: Mask-RCNN [32], Mask2Former [37], RTMDet-Ins-x [71], and ConvNeXt-V2 [35], as well as nine partially supervised methods, including three semi-supervised methods: PAIS [39], GuidedDistillation [11], and NoisyBoundaries [41]; and six weakly supervised methods: BoxInst [72], Discobox [73], BoxSnake [45], Apro_box2mask [74], Box2Mask [43], and RSPrompter [49]. Supervised methods use both detection and segmentation labels, semi-supervised methods use the same labels with additional unlabeled images, and weakly supervised methods use only detection labels. We compared the model accuracy trained on 4% and 7% of the training set to assess the model's learning capability under limited annotations. The detailed results are presented in Tables I and II.

A. Performance Comparison Based on 4% Training Set

Table I presents the accuracy evaluation results of the model trained on 4% ISAID data. The results demonstrate that the proposed RL-ISegNet has significant advantages. Compared to the ConvNeXt-V2, our detection results show an average improvement of +7.2% in AP, AP_{50}, and AP_{75}. In the segmentation results, there is an average increase of +7.0% in the three metrics. When compared to Box2Mask (ResNet-101), our detection results show an average improvement of +15.6% across three metrics, while our segmentation results show an average increase of +19.5%. Considering that the ResNet-101 backbone has nearly half the parameters of the ConvNeXt-V2-B backbone (62.9M versus 110.6M), the differences between Box2Mask (ResNet-101) and our proposed RL-ISegNet are significant. Therefore, we compared our RL-ISegNet with BoxSnake (Swin-L), which has 217.5M parameters, showing an average improvement of +4.9% in our detection results and +12.6% in our segmentation results. Overall, our proposed method shows an average improvement of +7.1% over the ConvNeXt-V2 baseline and +8.7% over BoxSnake (Swin-L). In summary, these results indicate that under the same number of training iterations (we achieve this by sampling the same samples five times, whereas the comparison methods repeat learning by increasing the epochs fivefold), merely increasing the number of training epochs is not effective in enhancing the model's learning efficiency on limited samples. In contrast, our proposed RL-ISegNet effectively improves sample utilization by continuously seeking the maximum reward value in a sequence to optimize model parameters.

B. Performance Comparison Based on 7% Training Set

Table II presents the accuracy evaluation results of the model trained on 7% ISAID data. Compared to RTMDet-Ins-x, the average improvements in our detection metrics AP, AP_{50}, and AP_{75} are +6.1%, while for segmentation, the improvements are +9.7%. When compared to ConvNeXt-V2, the three metrics show average improvements of +6.2% in our detection results and +5.1% in our segmentation results. In comparison to BoxSnake (Swin-L), the average improvements in our detection and our segmentation results are +5.8% and +14.1%, respectively. When compared to Box2Mask (Swin-L), the average improvements are +11.5% in our detection results and +16.2% in our segmentation results. Overall, our proposed method outperforms the second-best algorithm (ConvNeXt-V2) by an average of +5.7%, which is a decrease from the 4% results (+7.1%). From the above results, it can be observed that our method shows more significant advantages in the 4%

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TABLE II

COMPARISON OF INSTANCE SEGMENTATION MODEL ACCURACY (7% ISAID TRAINING SET). BOLD INDICATES THE BEST RESULTS AND UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS IN THE TABLE

Method	Backbone	#Params	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg (%)	FLOPs	FPS
Mask-RCNN	Resnet-101	63.1M	27.3	43.8	29.2	22.1	39.6	21.3	30.6	134G	45.9
Mask2Former	Resnet-101	63.0M	26.9	42.7	28.0	26.4	42.7	27.7	32.4	85.6G	23.5
RTMDet-Ins-x	CSPNeXt	102.7M	39.4	60.0	42.5	30.2	53.1	28.3	42.3	183G	31.4
ConvNeXt-V2	ConvNeXt-V2-B	110.6M	<u>39.5</u>	58.3	<u>44.0</u>	<u>34.2</u>	<u>55.5</u>	<u>35.6</u>	<u>44.5</u>	172G	25.1
PAIS	Resnet-101	56.4M	18.9	36.4	17.0	16.0	32.2	14.1	22.4	69.4G	5.9
GuidedDistillation	Resnet-101	62.9M	19.3	33.2	19.4	22.8	40.2	22.4	26.2	171.3G	13.7
Noisyboundaries	Resnet-101	62.82M	25.0	43.9	25.0	21.3	40.2	19.7	29.2	134.6G	28.3
BoxInst	Resnet-101	53.98M	28.1	45.8	29.2	13.2	30.6	9.7	26.1	115G	24.9
Discobox	Resnet-101	65.06M	19.9	35.0	19.6	18.3	35.0	17.3	24.2	124.7G	29.6
Apro_box2mask	Resnet-101	62.89M	24.7	43.7	23.7	9.9	26.4	5.7	22.4	78.5G	15.9
BoxSnake	Resnet-101	45.9M	33.5	53.1	36.0	23.2	43.1	21.1	35.0	136.5G	19.7
Box2Mask	Resnet-101	62.9M	30.8	50.2	31.0	20.2	40.8	17.7	31.8	78.9G	15.5
RSPrompter	ViT-L	320.2M	31.7	48.6	34.5	28.2	47.5	28.3	36.5	664.5G	6.5
BoxSnake	Swin-L	217.5M	38.3	<u>64.0</u>	40.7	25.8	48.4	24.3	40.3	607.1G	8.6
Box2Mask	Swin-L	215.5M	34.6	56.3	35.0	24.0	46.2	21.9	36.3	234.4G	13.6
Ours	ConvNeXt-V2-B	110.6M	**45.0**	**65.7**	**49.6**	**38.4**	**62.5**	**39.8**	**50.2**	172G	25.1

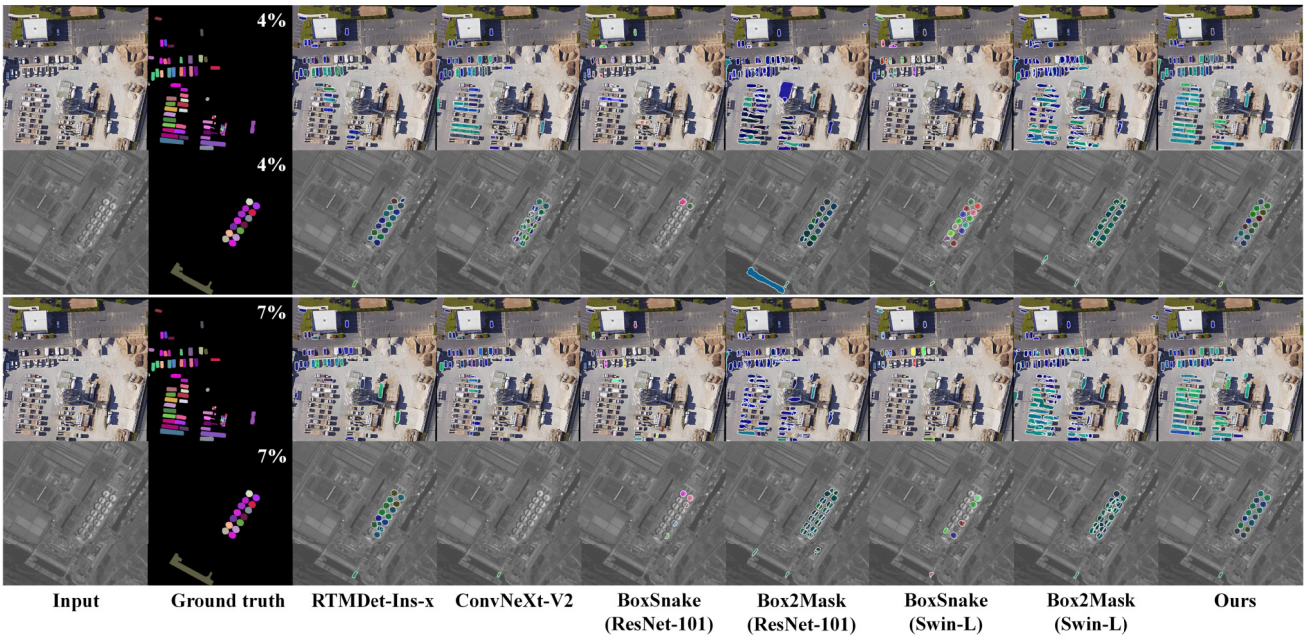


Fig. 8. Visualization results of instance segmentation. These models are trained on 4% (top) and 7% (bottom) ISAID data.

training set, indicating that the algorithm proposed in this article is effective in learning with limited samples.

C. Visualization Results

In Fig. 8, the results for vehicles and storage_tanks show relatively poor performance for other methods. For the vehicles, our algorithm has further improved the recall rate for vehicles, while other methods fail to effectively segment vehicles in both the 4% and 7% results. This is primarily due to vehicles being obscured by rocks, causing confusion between the vehicles and the background, making them difficult to identify. This indicates that our algorithm has effectively learned to handle challenging samples. In the results for storage_tanks, ConvNeXt-V2 shows missed detections of storage_tanks, suggesting that the model has developed a bias and has overlooked tank features. The missed detection of pools in the 4% results and storage_tanks in the 7% results illustrates a common issue where data imbalance in limited samples leads to model bias. Our algorithm addresses this issue by detecting these targets to achieve higher rewards, thereby enhancing the learning of these objects and reducing the occurrence of such problems.

VI. DISCUSSION

A. Evaluation of Data Efficacy

To intuitively understand the improvement in data efficiency, we compared our method with ConvNeXt-V2 trained on 100% of the training data. Specifically, ConvNeXt-V2 (100%) has AP, AP_{50}, and AP_{75} values of 47.4%, 68.0%, and 53.2% for detection results, and 40.9%, 64.9%, and 43.2% for segmentation results. The average of AP_{box} and AP_{mask} for ConvNeXt-V2 (100%) is 44.2%. In contrast, our proposed RL-ISegNet method achieves average AP values of 38.9%

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TABLE III

ROBUSTNESS ANALYSIS OF INSTANCE SEGMENTATION MODEL. BOLD INDICATES THE OPTIMAL AVERAGE VALUE FOR EACH TRANSFORMATION. S REPRESENTS THE SEVERITY, WITH A TOTAL OF ONE–FIVE LEVELS, AND 0 REMAINS UNCHANGED

Method	Transform	s=0	s=1	s=2	s=3	s=4	s=5	avg
ConvNeXt-V2 (7%)	Brightness	36.9	32.8	30.9	28.2	24.3	18.8	27.0
	Frost	–	22.7	15.4	10.7	10.3	8.3	13.5
	Fog	–	23.9	21.0	18.6	17.7	14.6	19.2
	Snow	–	18.6	11.0	8.2	3.3	4.5	9.1
	Shot noise	–	15.7	7.3	2.3	0.3	0.1	5.1
	Contrast	–	27.1	22.8	16.5	6.9	1.8	15.0
ConvNeXt-V2 (100%)	Brightness	44.2	43.5	41.4	37.6	32.1	24.7	35.9
	Frost	–	31.8	22.3	16.9	16.4	14.3	20.2
	Fog	–	33.8	29.4	26.5	25.9	23.0	27.7
	Snow	–	25.6	15.1	13.8	7.9	6.7	**13.8**
	Shot noise	–	15.0	5.5	1.2	0.3	0.1	4.4
	Contrast	–	38.7	34.2	26.7	14.2	6.0	24.0
Ours (7%)	Brightness	41.7	41.4	40.4	38.1	35.0	31.4	**37.3**
	Frost	–	32.4	24.6	20.0	19.7	17.2	**22.8**
	Fog	–	34.6	32.3	30.1	29.1	24.8	**30.2**
	Snow	–	20.7	15.2	10.8	7.1	10.6	12.9
	Shot noise	–	28.7	23.0	16.8	9.7	6.0	**16.8**
	Contrast	–	38.2	35.6	30.6	17.7	4.8	**25.4**

and 41.7% for the 4% and 7% training sets, respectively, while ConvNeXt-V2 achieves average AP values of 32.9% and 36.9% for the same training set sizes. The results indicate that the ConvNeXt-V2 (4%) model demonstrates 74.4% of the performance of the ConvNeXt-V2 (100%) model, whereas our RL-ISegNet (4%) achieves 88.0% of the performance of the ConvNeXt-V2 (100%) model, reflecting an improvement in data efficiency of 13.6%. Similarly, the ConvNeXt-V2 (7%) model achieves 83.5% of the performance of the ConvNeXt-V2 (100%) model, while our RL-ISegNet (7%) achieves 94.3% of the performance, resulting in a data efficiency improvement of 10.8%. Overall, our proposed RL-ISegNet method demonstrates an average data efficiency improvement of +12.2%. Moreover, it reaches 94.3% accuracy of the ConvNeXt-V2 (100%) model while utilizing only 7% of the training data. This illustrates that the method presented in this article significantly enhances data mining capabilities under limited sample conditions, providing a notable improvement over the ConvNeXt-V2 model.

B. Robustness Analysis

To further understand the robustness of the proposed method, we simulated the diversity of remote sensing images by applying transformations (adjusting brightness, contrast, and adding snow, fog, frost, noise, etc.) and directly evaluated the accuracy using the previously trained models. As shown in Table III, we compared three methods: ConvNeXt-V2 (7%), ConvNeXt-V2 (100%), and our RL-ISegNet (7%). The results indicate that our RL-ISegNet (7%) method outperformed ConvNeXt-V2 (7%) across six transformations (brightness, frost, fog, snow, shot noise, and contrast) by +10.3%, +9.3%, +11.0%, +3.8%, +11.7%, and +10.4%, respectively. Compared to ConvNeXt-V2 (100%), our RL-ISegNet (7%) method performed slightly worse only in the snow transformation (−0.9%), while outperforming it in the other transformations by +1.4%, +2.6%, +2.5%, +12.4%, and +1.4%. Overall, our proposed model demonstrates significantly greater robustness than the ConvNeXt-V2 (7%) model and exhibits better noise resilience than ConvNeXt-V2 (100%). This indicates that the algorithm presented in this article outperforms comparison methods in generalization performance, suggesting that the features we have derived are more universal and uncover common representations of target objects.

C. Analysis of the RPN Head Optimization

To understand the changes in ROI proposals generated by the RPN head before and after RLs, we plotted a scatter density map of the IOU and classification scores for the ROI proposals. This figure illustrates the quality of 70 683 ROI proposals corresponding to real instances. As shown in Fig. 9, a high density in the top-right (TR) area indicates better performance of the model's RPN head. We observe that some ROI proposals from ConvNeXt-V2 (7%), ConvNeXt-V2 (100%), and our RL-ISegNet (7%) concentrate in this area, with our method (TR:32.2%) exhibiting a higher density than the other two. The top-left (TL) area indicates that ROI proposals have a small IOU but likely contain instance targets. This area typically includes proposals that can localize the target but fail to accurately capture its size. Among the three methods, ConvNeXt-V2 (100%) performs the worst (TL: 5.2%) in this region. The bottom-left (BL) area shows poorer quality ROI proposals, where ConvNeXt-V2 (7%) has higher density (BL: 62.4%). The bottom-right (BR) area indicates that the generated ROI proposals localize the targets well but classify them as background, with ConvNeXt-V2 (7%) showing significantly more proposals in this area compared to ConvNeXt-V2 (100%) and our RL-ISegNet (7%) method. Overall, our RL-ISegNet (7%) method, with the RPN head optimized through RL, clearly outperforms ConvNeXt-V2 (7%) and demonstrates performance similar to ConvNeXt-V2 (100%). This suggests that our RL approach for the RPN head is effective in improving the quality of ROI proposals generated under limited samples.

D. Analysis of the Detection Head Optimization

To understand the changes in the detection boxes generated by the detection head before and after RLs, we plotted a scatter density map of the IOU and classification scores for the detection boxes. This figure illustrates the quality of 70 683 detection boxes corresponding to real instances. As shown in Fig. 10, a higher density in the TR area indicates better performance of the detection head in detecting targets. From the figure, we can see that our RL-ISegNet (7%) method has a higher density (TR: 21.2%) in this area compared to ConvNeXt-V2 (7%) and ConvNeXt-V2 (100%). Overall, our RL-ISegNet (7%) method, with the detection head optimized through RL, significantly outperforms ConvNeXt-V2 (7%) and slightly exceeds the performance of ConvNeXt-V2 (100%). This indicates that our RL approach for the detection head is effective in improving the quality of detection boxes generated under limited samples.

E. Analysis of the Segmentation Head Optimization

To understand the performance changes of the segmentation head before and after RLs, we plotted t-distributed

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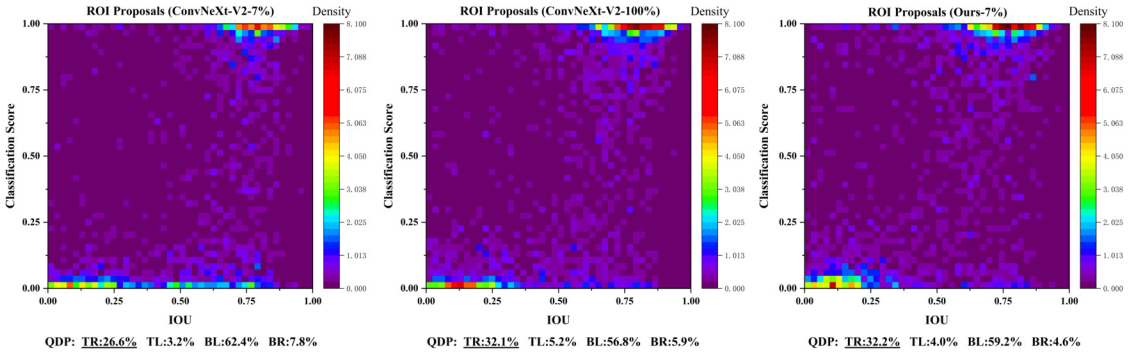


Fig. 9. Scatter density plot of ROI proposals. The horizontal axis represents the IOU, indicating the overlap between the predicted ROI proposal and the ground-truth box, while the vertical axis shows the classification score, reflecting the model's confidence that the ROI proposal corresponds to an instance target. Quadrant distribution percentage (QDP) denotes the proportion of object samples located in each quadrant of the score-IOU space. TR (score ≥ 0.5 and IOU ≥ 0.5), TL (score ≥ 0.5 and IOU <0.5), BR (score <0.5 and IOU ≥ 0.5), and BL (score <0.5 and IOU <0.5).

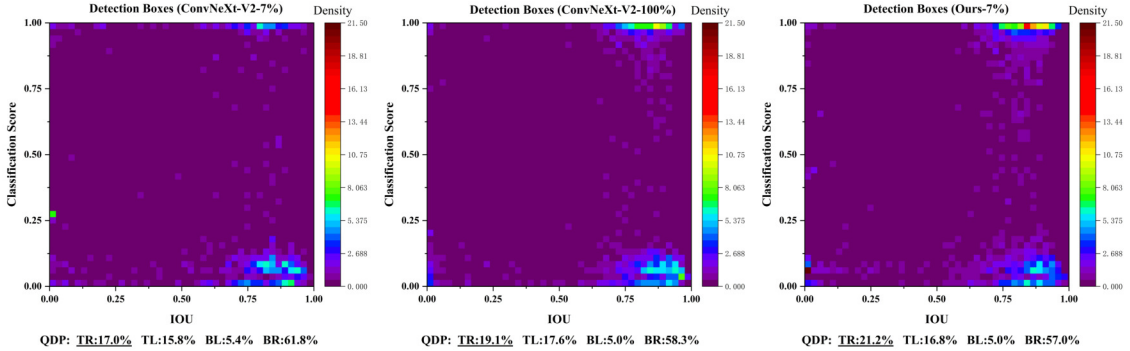


Fig. 10. Scatter density plot of detection boxes. The horizontal axis represents the IOU, indicating the overlap between the predicted detection boxes and the ground-truth boxes, while the vertical axis shows the classification score, reflecting the confidence in the true class of the detection boxes.

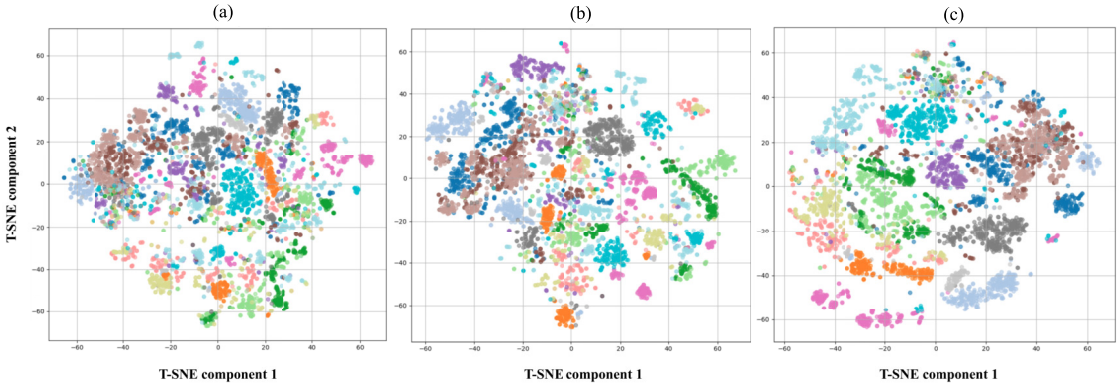


Fig. 11. t-SNE visualization of the segmentation head. Each point represents a feature vector, and each color represents a category. (a) t-SNE visualization (ConvNeXt-V2-7%). (b) t-SNE visualization (ConvNeXt-V2-100%). (c) t-SNE visualization (ours-7%).

stochastic neighbor embedding (t-SNE) visualizations for the segmentation heads of three models: ConvNeXt-V2 (7%), ConvNeXt-V2 (100%), and RL-ISegNet (7%). Fig. 11(a) shows the results of the ConvNeXt-V2 (7%) model. Although there is some clustering, the separation between many categories is low. This indicates that when only 7% of the data is used, the model's discriminative ability in the feature space is limited, making it difficult to fully capture the differences between categories. Compared to the ConvNeXt-V2 (7%), the ConvNeXt-V2 (100%) model trained with 100% of the data shows significant improvement in Fig. 11(b). The separation between categories is better, the clustering is tighter, and interclass overlap is reduced. This suggests that a sufficient amount of data helps the model learn the features of each category more effectively, enabling more accurate classification in the feature space. In Fig. 11(c), compared to the ConvNeXt-V2 (7%), the clustering effect of our proposed RL-ISegNet (7%) method is closer to that of ConvNeXt-V2 (100%), with improved separation between most categories and tighter clustering, demonstrating that RL-ISegNet has better generalization capability under limited data conditions. Overall, these results show that RL-ISegNet,

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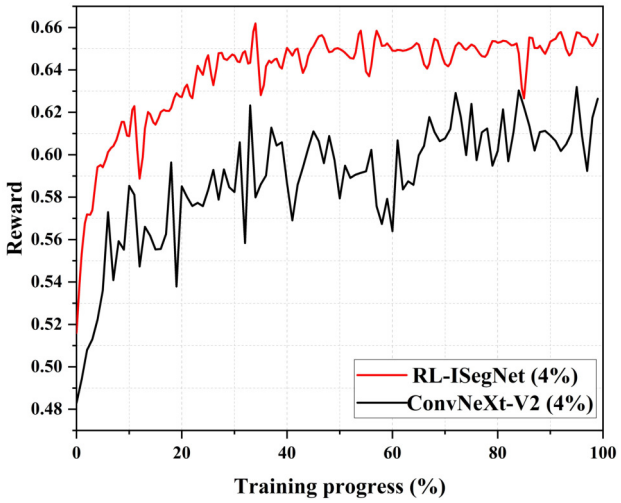


Fig. 12. Reward progress curve. The trend of reward values changing with training progress in the RL process. In the figure, the red and black curves represent the changes in reward values in two different states, before and after RL optimizations.

through RL and other optimization strategies, can effectively distinguish different categories with limited data, maintaining good interclass distribution and intraclass consistency.

F. Reward Progression

To understand the changes in rewards before and after RL optimizations, we plotted the average rewards (RPN, detection, and segmentation) for ConvNeXt-V2 (4%) and our RL-ISegNet (4%) at different stages of training. As shown in Fig. 12, the rewards of our RL-ISegNet (4%) are generally higher than those of ConvNeXt-V2 (4%), indicating that the RL optimization leads to a continuous increase in cumulative rewards and a certain improvement in model performance. In addition, as shown in Fig. 13, we visualized the model's ability to learn information at different training stages for ConvNeXt-V2 (4%) and our RL-ISegNet (4%). From the results, it is evident that in the early stages of training, our model demonstrates significantly better learning ability for vehicles compared to ConvNeXt-V2 (4%). However, in the later stages, ConvNeXt-V2 (4%) still fails to recognize some small cars, while our RL-ISegNet (4%) can segment these more challenging objects to some extent. Overall, our RL-ISegNet (4%) enhances information extraction through RL, improving its inductive ability with limited information.

G. Ablation Experiment

In this section, we conduct a series of ablation experiments on the ISAID dataset to verify the effectiveness of our proposed method.

1) Impact of Different Modules: In this section, we examine the impact of ablating different modules on model accuracy (as shown in Table IV). In Table IV, we first apply RL to the RPN head, detection head, and segmentation head. The results show improvements of +3.6%, +2.2%, and +4.2% over the baseline, respectively. When optimizing all three components together (RPN, Det, and Seg), the accuracy increases by

TABLE IV

ABLATION EXPERIMENT (4% ISAID TRAINING SET). RPN, DET, AND SEG REPRESENT THE RL APPLIED TO THE RPN HEAD, DETECTION HEAD, AND SEGMENTATION HEAD, RESPECTIVELY. CONS REFERS TO THE TASK CONSISTENCY CONSTRAINT, WHILE FISHER DENOTES THE PARAMETER FREEZING BASED ON FISHER INFORMATION

RPN	Det	Seg	Cons	Fisher	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg
					36.1	53.2	40.8	29.7	49.2	30.5	39.9
✓					38.0	59.9	41.9	32.6	55.6	33.2	43.5
	✓				36.6	57.6	40.8	31.7	53.6	32.0	42.1
		✓			38.5	59.4	42.7	33.5	55.7	34.7	44.1
✓	✓	✓			40.2	60.5	44.9	34.0	56.6	35.0	45.2
✓	✓	✓	✓	✓	39.8	61.7	44.1	34.1	57.9	34.1	45.3
✓	✓	✓	✓		42.3	62.3	47.0	35.4	58.5	36.6	47.0

TABLE V

MODEL ACCURACY UNDER DIFFERENT TRAINING SET PROPORTIONS

Data ratio	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg
Ours(1%)	35.1	52.4	38.7	29.8	49.3	30.7	39.3
Ours(4%)	42.3	62.3	47.0	35.4	58.5	36.6	47.0
Ours(7%)	45.0	65.7	49.6	38.4	62.5	39.8	50.2
Ours(10%)	46.3	66.6	51.9	39.3	63.1	41.2	51.4
ConvNeXt-V2(100%)	47.4	68.0	53.2	40.9	64.9	43.2	52.9

+5.3% compared to the baseline. This indicates that our method effectively mines information from hard samples in limited data. Due to significant conflicts between tasks, we introduced consistency constraints for optimization. This improved accuracy from 45.2% to 47.0% (+1.8%). To reduce training time, we applied Fisher information to freeze some parameters; however, after freezing, accuracy decreased by 1.7% compared to 47.0%. Specifically, the Fisher information captures the global importance of parameters based on their sensitivity to model output, while training requires adjusting parameters driven by local, data-specific errors. This mismatch means that parameters selected for freezing may not align with those needing updates, and excessive freezing can hinder effective optimization, leading to performance degradation. Addressing this limitation will be one of our future research directions.

2) Impact of Different Training Set Ratios: As shown in Table V, training with only 1% of the data leads to a notable performance gap compared to full supervision (39.3% versus 52.9%), suggesting that minimal supervision is insufficient. Increasing the ratio to 4% yields a 7.7-point improvement, indicating that even limited additional data can significantly enhance performance. A further increase to 7% brings a 3.2-point gain (47.0%→ 50.2%), reflecting higher data efficiency than the subsequent 7%–10% interval, which yields only 1.2 points. Remarkably, using just 7% of the data, the model attains close to 95% of the full performance, highlighting it as a cost-effective tradeoff point.

3) Impact of Time Steps: In this section, we report the impact of different time steps in RL on model accuracy in Table VI. The results show a gradual accuracy increase from two to eight steps across various metrics, indicating that increasing the time step in RL yields better results. However, when the step is set to 5, the accuracy improvement begins to

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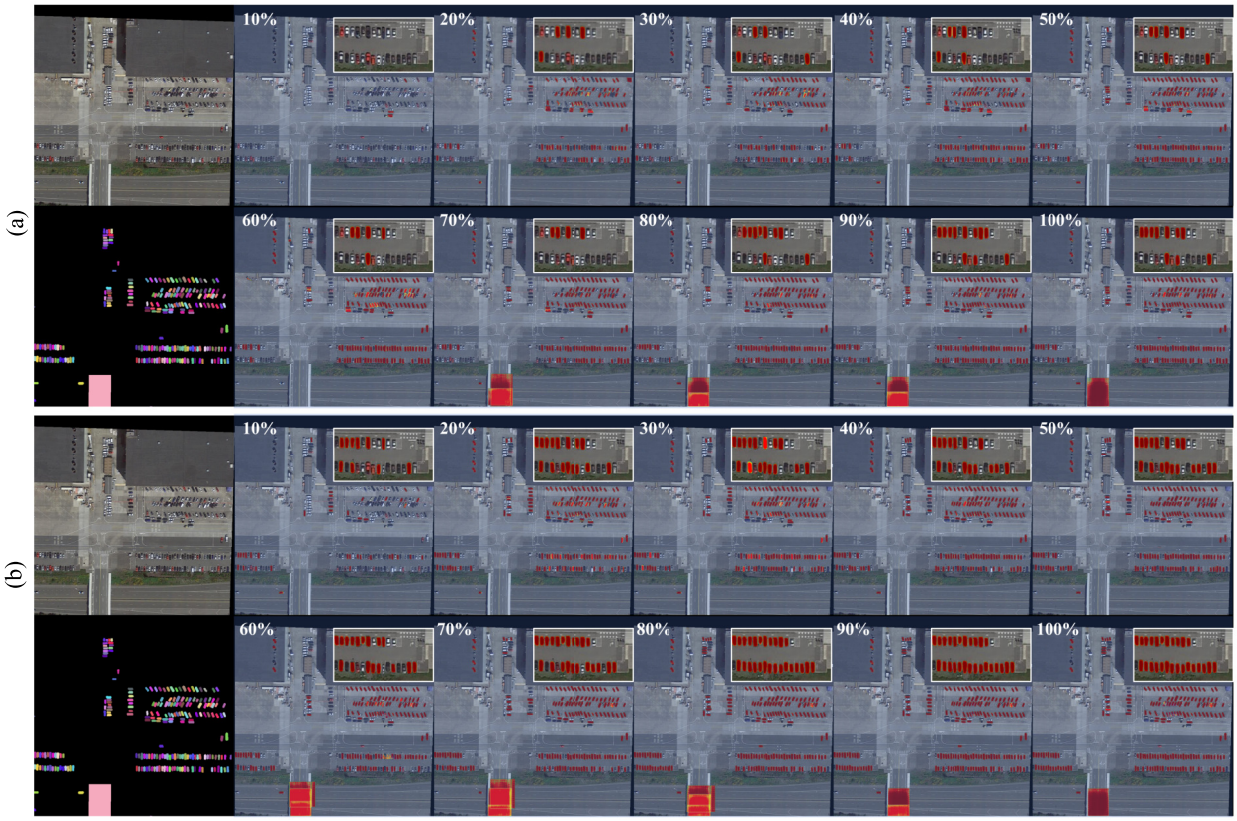


Fig. 13. Results under different training processes (from 10% to 100%). The red color in the figure represents the confidence level of the target object; the redder the color, the higher the confidence level, and vice versa.

TABLE VI
IMPACT OF TIME STEP ON ACCURACY (4% ISAID TRAINING SET)

Step	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg
2	37.7	59.4	40.8	32.1	56.0	32.1	43.0
3	40.9	61.6	45.3	34.8	58.1	36.1	46.1
4	41.8	62.5	46.2	35.1	58.7	35.7	46.7
5	42.3	62.3	47.0	35.4	58.5	36.6	47.0
6	42.0	62.5	46.5	35.2	58.8	36.9	47.0
7	42.4	62.4	46.9	35.6	58.6	36.7	47.1
8	42.6	62.6	47.4	35.8	58.8	37.7	47.5

TABLE VII
IMPACT OF CONSISTENCY CONSTRAINT ON INTERTASK CONFLICT (E.G., RPN-DET INDICATES THE INFLUENCE OF RPN ON DET). THE NIR REPRESENTS THE PROPORTION OF NEGATIVE INFLUENCE OCCURRENCES THROUGHOUT THE ITERATIONS. THE MPI DENOTES THE AVERAGE POSITIVE INFLUENCE VALUE, WHILE THE MNI DENOTES THE AVERAGE NEGATIVE INFLUENCE VALUE

Influence	NIR	mPI	nNI
RPN-DET	48.55%→30.69%	0.1233→0.1599	-0.1642→-0.1346
RPN-SEG	50.03%→30.66%	0.0503→0.0810	-0.0566→-0.0511
DET-RPN	43.14%→35.16%	0.2510→0.2780	-1.2360→-0.5675
DET-SEG	47.50%→28.75%	0.0534→0.0908	-0.0533→-0.0513
SEG-RPN	42.21%→33.84%	0.2502→0.2974	-1.0980→-0.5132
SEG-DET	36.07%→27.41%	0.1475→0.1885	-0.1659→-0.1398

plateau, suggesting that beyond a certain step length, model performance saturates, and further gains diminish. Considering that the time cost increases with more steps, we chose to use a step of 5 for training in this study.

4) Intertask Consistency Analysis: To understand the changes in task conflicts after applying task consistency optimization, we calculated three metrics: NIR, mPI, and mNI, as shown in Table VII. The average intertask conflict indicator (NIR) was 44.58% before applying consistency constraints and decreased to 31.09%, reducing the conflict rate by 13.49% after applying consistency constraints. The average intertask conflict indicator (mPI) was 0.1460 before applying consistency constraints and increased to 0.1819, indicating a 24.59% increase in the degree of positive influence. The average intertask conflict indicator (mNI) was −0.4623 before applying consistency constraints and improved to −0.2429, reducing the degree of negative influence by 47.46%. In summary, these results indicate that consistency optimization effectively reduced intertask conflicts, with a 13.49% decrease in conflict occurrences, a 47.46% reduction in the degree of negative influence, and a 24.59% increase in the degree of positive influence.

5) Effectiveness and Training Overhead of RL: As shown in Table VIII, incorporating RL consistently improves model accuracy across all training data ratios. The most significant gain is observed under the 4% data setting, where RL yields a 7.1-point increase in average accuracy (47.0% versus 39.9%), highlighting its effectiveness in low-supervision scenarios.

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TABLE VIII
IMPACT OF THE RL ON MODEL ACCURACY. TIME DENOTES THE TRAINING DURATION. W/O RL MEANS WITHOUT RL

Architectures design	Data ratio	Avg	Time
Ours	1%	39.3	11.1h
w/o RL	1%	34.5	5.4h
Ours	4%	47.0	28.3h
w/o RL	4%	39.9	12.7h
Ours	7%	50.2	42.3h
w/o RL	7%	44.5	21.8h
Ours	10%	51.4	57.2h
w/o RL	10%	46.5	28.8h

TABLE IX
IMPACT OF THRESHOLD $I(\theta_i)$ ON ACCURACY. TIME DENOTES THE TRAINING DURATION

ϵ	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg	Time
0.005	34.3	53.9	36.3	29.7	51.5	31.3	39.5	21.8h
0.00005	38.6	60.0	42.7	33.8	56.0	34.7	44.3	22.5h
0.0000005	39.8	61.7	44.1	34.1	57.9	34.1	45.3	24.2h
–	42.3	62.3	47.0	35.4	58.5	36.6	47.0	28.3h

However, this performance improvement comes at the cost of substantially increased training time. In all cases, the RL models require nearly twice the training duration compared to their non-RL counterparts. These results indicate that while RL boosts performance, its computational overhead needs to be considered.

6) Time Consumption: In this section, we report the training time and model accuracy under different Fisher information thresholds in Table IX. When the ε threshold is $5e^{-3}$, our model's accuracy (39.5%) is close to the ConvNeXt-V2 baseline (39.9%), while the training time is reduced by 22.97% compared to training without parameter freezing. With an ε threshold of $5e^{-5}$, our model achieves an accuracy of 44.3%, with a reduction in both training time and accuracy by 20.49% and 2.7%, respectively, compared to no freezing. When the ε threshold is $5e^{-7}$, the model accuracy is 45.3%, with training time reduced by 14.49% and accuracy dropping by 1.7%. These results show that our method achieves limited optimization in reducing training time while maintaining accuracy.

Although our method does not achieve significant improvements in training efficiency, it is important to note that the RL framework only affects the training phase and does not increase inference cost. As shown in Table I, the inference speed (FPS) and computational complexity (FLOPs) of our method are consistent with the ConvNeXt-V2-B baseline, indicating no additional delay during testing. Compared to heavier backbones such as ViT-L and Swin-L, our method achieves clear advantages in both accuracy and inference efficiency (FPS and FLOPs). Therefore, our method remains practical and efficient for deployment. In summary, this study prioritizes accuracy over training efficiency, aiming to achieve the best possible performance under limited supervision. Nevertheless, for scenarios where training cost is a key concern, we recommend using a threshold of $\varepsilon = 5e^{-7}$.

H. Limitations and Outlook

Specifically, the integration of RL introduces substantial computational overhead, significantly increasing training time. Although we propose a parameter freezing strategy based on Fisher information to mitigate this issue, our results show that reducing trainable parameters leads to a tradeoff: shorter training time is often accompanied by decreased accuracy. Inspired by recent advances such as the low-rank adaptation (LoRA) series [75], future work could explore low-rank adaptation techniques to address this challenge. Instead of directly reducing parameters based on importance scores, a LoRA-style framework introduces lightweight trainable adapters while freezing the majority of the base model. RL could, then, focus exclusively on optimizing these compact adapters, which are later combined with the frozen parameters. Unlike Fisher information, this residual-style parameterization (LoRA) may better preserve model accuracy while significantly reducing training cost, thus offering a more stable tradeoff between efficiency and performance. In addition, our RL framework currently does not support semi-supervised or weakly supervised framework, as it requires both detection and segmentation labels. In semi-supervised learning, the challenge lies in designing reward functions for unlabeled data. In weakly supervised learning, we must design reward functions for the segmentation head without segmentation labels. These challenges will be the focus of our future work.

VII. CONCLUSION

In this article, we focus on improving data utilization efficiency with limited samples to reduce the reliance on large amounts of labeled data. We propose an RL-driven instance segmentation framework, RL-ISegNet, which uses policy gradient optimization to enhance the learning of hard samples in limited labeled data, thereby improving data utilization efficiency. Specifically, RL-ISegNet treats the RPN head, detection head, and segmentation head of instance segmentation as RL policies, using cumulative rewards from each subtask to provide feedback for optimizing current network parameters, aiming to find optimal task network parameters. To reduce subtask conflicts and the time cost of sequential learning during RL, we designed a metric to measure subtask conflicts, increasing the sampling probability of trajectories with positive impacts for each subtask. Additionally, we developed a Fisher information-based parameter freezing method to reduce training time. The results show that our method improves data utilization efficiency by 12.4%. On 4% and 7% of the training dataset, accuracy increased by 7.1% and 5.7%, respectively. The RL-ISegNet model trained on 7% of the dataset achieved 94.3% of the accuracy of the baseline model trained on 100% of the dataset. In terms of task conflicts, the conflict rate decreased by 13.49%, the degree of negative impacts decreased by 47.46%, and the degree of positive impacts increased by 24.59%.

APPENDIX
EXPERIMENTAL RESULTS ON EXTENDED DATASETS

To further validate the effectiveness and generalizability of the proposed method, we conduct additional evaluations on

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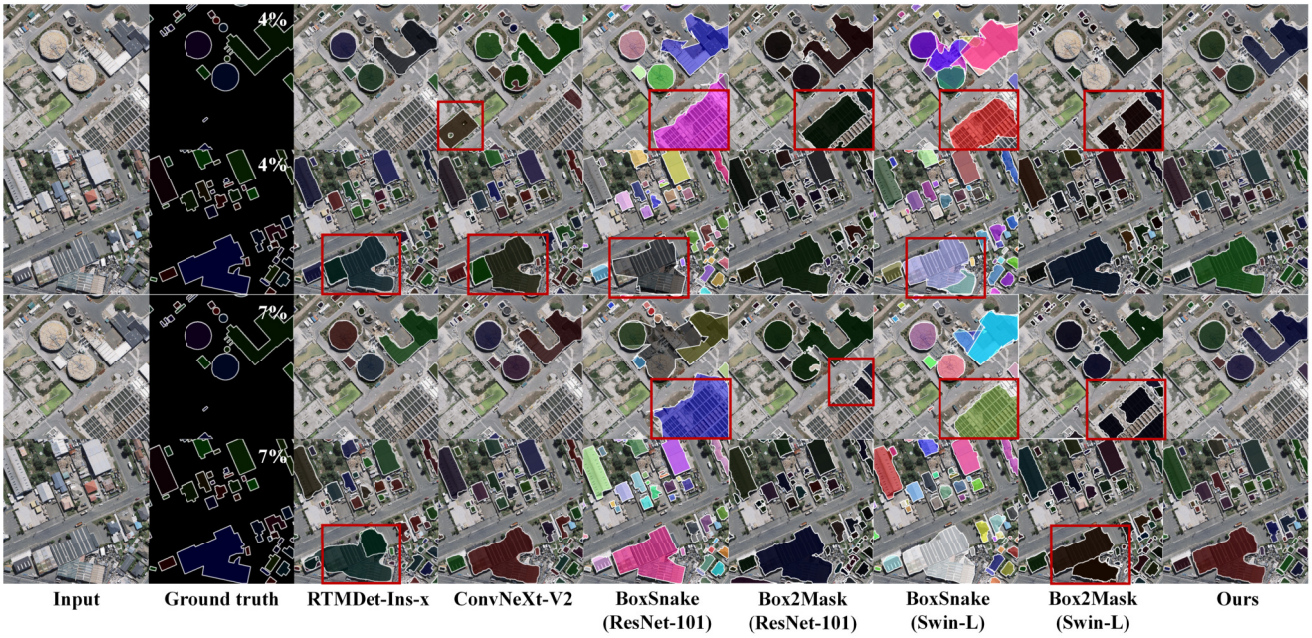


Fig. 14. Visualization results of instance segmentation. These models are trained on 4% (top) and 7% (bottom) WHU data.

TABLE X

COMPARISON OF INSTANCE SEGMENTATION MODEL ACCURACY (4% WHU TRAINING SET). BOLD INDICATES THE BEST RESULTS AND UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS IN THE TABLE

Method	Backbone	#Params	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg (%)	FLOPs	FPS
Mask-RCNN	Resnet-101	63.1M	48.6	74.8	55.6	49.8	75.9	57.9	60.4	134G	25.4
Mask2Former	Resnet-101	63.0M	36.3	55.2	40.2	43.8	70.1	49.0	49.1	85.6G	25.2
RTMDet-Ins-x	CSPNeXt	102.7M	57.4	81.1	65.7	57.0	82.3	65.8	68.2	183G	24.9
ConvNeXt-V2	ConvNeXt-V2-B	110.6M	63.1	82.2	72.8	61.9	83.1	73.1	72.7	172G	25.3
PAIS	Resnet-101	56.4M	51.7	75.0	57.9	50.2	73.8	56.3	60.8	69.4G	8.3
GuidedDistillation	Resnet-101	62.9M	50.2	72.9	57.7	53.9	76.7	62.4	62.3	171.3G	14.6
Noisyboundaries	Resnet-101	62.82M	49.8	77.6	56.4	50.6	78.7	59.2	62.1	134.6G	31.4
BoxInst	Resnet-101	53.98M	54.0	79.0	60.6	36.6	76.7	29.5	56.1	115G	26.0
Discobox	Resnet-101	65.06M	45.1	69.5	49.2	47.8	77.0	53.6	57.0	124.7G	35.8
Apro_box2mask	Resnet-101	62.89M	47.7	74.9	52.8	46.1	76.6	50.9	58.2	78.5G	18.9
BoxSnake	Resnet-101	45.9M	62.4	84.9	71.8	59.0	85.0	69.3	72.1	136.6G	23.4
Box2Mask	Resnet-101	62.9M	56.3	82.7	63.1	52.1	83.2	59.2	66.1	78.9G	18.8
Rsprompter	ViT-L	320.2M	57.6	84.8	67.5	24.3	41.6	25.5	50.2	664.5G	24.4
BoxSnake	Swin-L	217.5M	64.3	87.1	75.2	56.8	87.1	68.1	73.1	607.1G	13.4
Box2Mask	Swin-L	215.5M	56.2	81.3	63.1	52.7	82.5	60.1	66.0	234.4G	15.8
Ours	ConvNeXt-V2-B	110.6M	**69.1**	**88.7**	**79.8**	**68.0**	**89.1**	**79.9**	**79.1**	172G	25.3

two datasets, including the WHU building extraction dataset [76] and the NWPU VHR-10 dataset [77].

A. WHU

We use data from the WHU building extraction dataset. This dataset consists of 8188 nonoverlapping RGB images, each with a resolution of 512 × 512 pixels, and contains a single building category. The images were captured in Christchurch, New Zealand, with spatial resolutions ranging from 0.0075 to 0.3 m. In this article, 4736 images were assigned to the training set (from which we randomly constructed 4% (190 images) and 7% (332 images) training subsets) and 2416 images were assigned to the test set. To obtain instance-level annotations, we employed the connected component analysis module in OpenCV to convert semantic segmentation masks into instance segmentation format.

B. NWPU

The NWPU dataset is a remote sensing object detection dataset that includes ten object categories. It consists of 715 optical remote sensing images from Google Earth with spatial resolutions ranging from 0.5 to 2 m, and 85 pan-sharpened color infrared images from the Vaihingen dataset with a spatial resolution of 0.08 m. We allocated 640 images to the training set (from which we randomly constructed 4% (27 images) and 7% (45 images) training subsets) and 160 images to the test set. During training, all images were resized to 512 × 512. The instance annotations provided in [78] were used for training and evaluation.

1) Results of WHU Datasets: As shown in Tables X and XI, our proposed method consistently achieves the best performance under both 4% and 7% training data settings on the WHU dataset. Specifically, our method outperforms

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TABLE XI

COMPARISON OF INSTANCE SEGMENTATION MODEL ACCURACY (7% WHU TRAINING SET). BOLD INDICATES THE BEST RESULTS AND UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS IN THE TABLE

Method	Backbone	#Params	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg (%)	FLOPs	FPS
Mask-RCNN	Resnet-101	63.1M	51.0	76.0	58.9	51.9	76.3	61.1	62.5	134G	25.4
Mask2Former	Resnet-101	63.0M	45.0	68.3	50.6	33.4	60.0	26.0	47.2	85.6G	25.2
RTMDet-Ins-x	CSPNeXt	102.7M	60.9	83.3	69.7	60.1	84.4	69.7	71.4	183G	24.9
ConvNeXt-V2	ConvNeXt-V2-B	110.6M	64.4	82.6	74.5	<u>62.6</u>	83.3	74.1	73.6	172G	25.3
PAIS	Resnet-101	56.4M	55.7	78.0	62.7	54.7	76.7	63.9	65.3	69.4G	8.3
GuidedDistillation	Resnet-101	62.9M	53.7	75.8	61.7	55.9	77.6	65.1	65.0	171.3G	14.6
Noisyboundaries	Resnet-101	62.82M	54.0	79.6	61.9	53.7	80.5	62.5	65.4	134.6G	31.4
BoxInst	Resnet-101	53.98M	57.3	81.4	64.0	41.6	79.6	40.4	60.7	115G	26.0
Discobox	Resnet-101	65.06M	45.6	70.0	50.0	49.5	78.6	56.3	58.3	124.7G	35.8
Apro_box2mask	Resnet-101	62.89M	50.5	77.2	55.8	47.8	78.6	53.3	60.5	78.5G	18.9
BoxSnake	Resnet-101	45.9M	64.3	87.4	74.1	61.2	86.8	72.3	74.4	136.6G	23.4
Box2Mask	Resnet-101	62.9M	57.8	83.4	65.1	53.3	84.0	61.1	67.5	78.9G	18.8
RSPrompter	ViT-L	320.2M	55.9	83.5	66.8	41.0	65.0	46.6	59.8	664.5G	24.4
BoxSnake	Swin-L	217.5M	<u>65.5</u>	<u>88.8</u>	<u>76.4</u>	58.0	<u>87.9</u>	<u>74.3</u>	<u>75.2</u>	607.1G	13.4
Box2Mask	Swin-L	215.5M	58.4	83.6	65.9	55.2	84.6	63.4	68.5	234.4G	15.8
Ours	ConvNeXt-V2-B	110.6M	**70.4**	**88.9**	**80.8**	**69.2**	**89.3**	**80.7**	**79.9**	172G	25.3

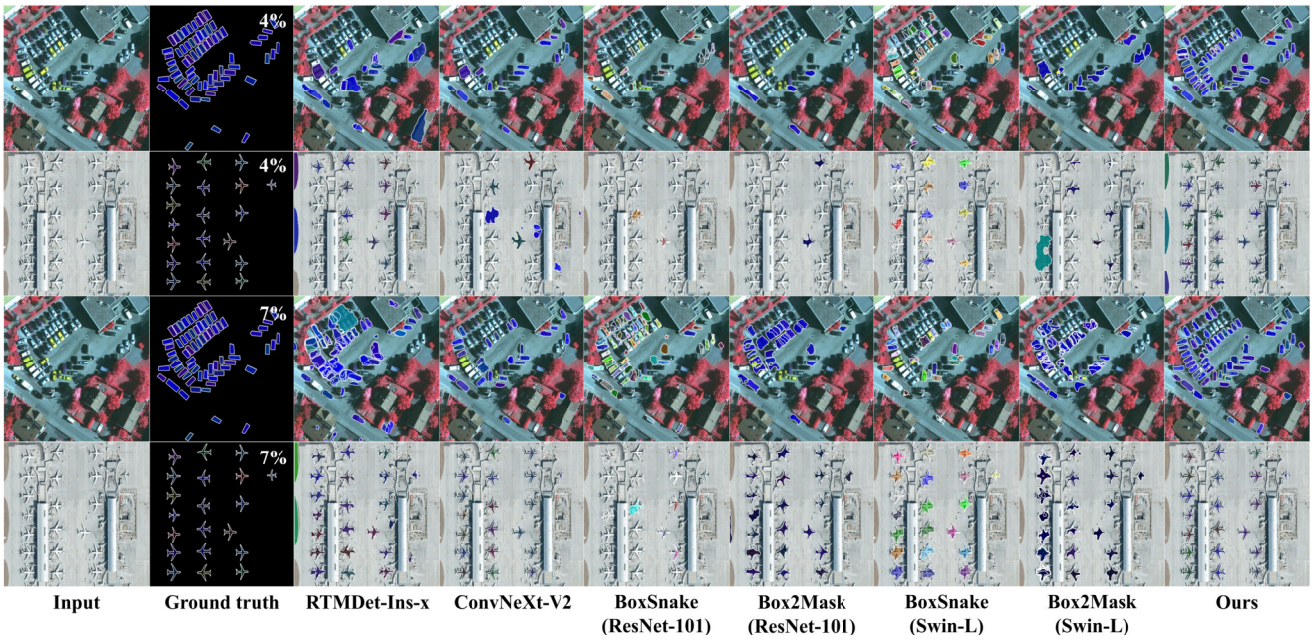


Fig. 15. Visualization results of instance segmentation. These models are trained on 4% (top) and 7% (bottom) NWPU data.

all baselines across box and mask AP metrics, with an average accuracy of 79.1% using only 4% of the training data, and further improving to 79.9% with 7%. Compared to other methods such as BoxSnake (Swin-L) and RTMDet-Ins-x, our model achieves significantly higher mask AP, particularly at stricter thresholds (e.g., AP^{75}_{mask}), indicating superior segmentation quality. As shown in Fig. 14, the compared methods exhibit a large number of false positives and over-segmentation issues when performing instance segmentation on buildings. Moreover, as shown in Tables I and X, we observe that semi-supervised methods exhibit significantly less stable performance in multiclass instance segmentation tasks compared to single-class scenarios. This is mainly due to the difficulty in ensuring the accuracy of pseudo-labels, as feature differences between categories make pseudo-labels more prone to class confusion and unclear boundaries, which, in turn, hampers the model's learning effectiveness.

2) Results of NWPU Datasets: As shown in Tables XII and XIII, our proposed method achieves the best performance under both 4% and 7% training data settings on the NWPU dataset. With only 4% of the training data, our model reaches an average accuracy of 60.0%, outperforming all baselines across both box and mask AP metrics. When the training data are increased to 7%, the performance further improves to 71.2%, achieving the highest scores in all AP metrics, including AP^{75}_{box} and AP^{75}_{mask}. Compared to strong baselines such as BoxSnake (Swin-L) and ConvNeXt-V2, our method shows superior performance, particularly in high-precision segmentation. As shown in Fig. 15, our method achieves the best performance in both object detection and segmentation, particularly demonstrating a clear advantage in aircraft

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TABLE XII

COMPARISON OF INSTANCE SEGMENTATION MODEL ACCURACY (4% NWPU TRAINING SET). BOLD INDICATES THE BEST RESULTS AND UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS IN THE

Method	Backbone	#Params	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg (%)	FLOPs	FPS
Mask-RCNN	Resnet-101	63.1M	20.6	34.4	22.7	16.4	31.1	15.7	23.5	134G	47.3
Mask2Former	Resnet-101	63.0M	35.9	56.1	38.2	37.1	59.4	38.1	44.1	85.6G	28.6
RTMDet-Ins-x	CSPNeXt	102.7M	40.6	63.0	44.6	24.0	39.9	26.6	39.8	183G	24.3
ConvNeXt-V2	ConvNeXt-V2-B	110.6M	47.5	70.6	53.1	43.3	68.1	44.0	54.4	172G	28.7
PAIS	Resnet-101	56.4M	39.5	65.2	38.3	35.1	57.5	34.6	45.0	69.4G	2.6
GuidedDistillation	Resnet-101	62.9M	27.1	39.0	30.2	40.6	62.4	42.5	40.3	171.3G	11.8
Noisyboundaries	Resnet-101	62.82M	33.7	55.1	37.7	31.6	51.8	35.0	40.8	134.6G	30.9
BoxInst	Resnet-101	53.98M	32.3	55.2	32.0	23.2	45.1	19.9	34.6	115G	29.1
Discobox	Resnet-101	65.06M	26.0	44.1	26.9	22.0	37.1	23.7	30.0	124.7G	30.3
Apro_box2mask	Resnet-101	62.89M	41.3	65.3	43.4	24.3	49.6	20.6	40.8	78.5G	16.3
BoxSnake	Resnet-101	45.9M	46.3	71.5	53.9	36.8	60.3	38.1	51.2	136.6G	16.1
Box2Mask	Resnet-101	62.9M	41.5	63.6	44.2	28.3	52.6	27.1	42.9	78.9G	18.1
Rsprompter	ViT-L	320.2M	34.5	62.4	32.8	12.1	20.0	10.8	28.8	664.5G	20.8
BoxSnake	Swin-L	217.5M	45.7	**80.6**	47.6	37.2	61.9	37.2	51.7	607.1G	15.7
Box2Mask	Swin-L	215.5M	39.7	62.0	42.0	28.8	51.8	26.4	41.8	234.4G	14.3
Ours	ConvNeXt-V2-B	110.6M	**51.3**	74.0	**59.3**	**48.9**	**73.2**	**53.0**	**60.0**	172G	28.7

TABLE XIII

COMPARISON OF INSTANCE SEGMENTATION MODEL ACCURACY (7% NWPU TRAINING SET). BOLD INDICATES THE BEST RESULTS AND UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS IN THE TABLE

Method	Backbone	#Params	AP_{box}	AP^{50}_{box}	AP^{75}_{box}	AP_{mask}	AP^{50}_{mask}	AP^{75}_{mask}	Avg (%)	FLOPs	FPS
Mask-RCNN	Resnet-101	63.1M	22.4	36.3	24.1	18.7	30.4	20.8	25.5	134G	47.3
Mask2Former	Resnet-101	63.0M	45.5	69.1	48.9	43.3	69.3	45.6	53.6	85.6G	28.6
RTMDet-Ins-x	CSPNeXt	102.7M	50.0	75.7	60.9	32.6	52.9	35.1	51.2	183G	24.3
ConvNeXt-V2	ConvNeXt-V2-B	110.6M	55.6	80.4	64.7	52.7	78.7	56.6	64.8	172G	28.7
PAIS	Resnet-101	56.4M	45.2	71.0	44.6	41.2	65.8	41.8	51.6	69.4G	2.6
GuidedDistillation	Resnet-101	62.9M	31.7	45.5	35.8	41.9	64.4	43.7	43.8	171.3G	11.8
Noisyboundaries	Resnet-101	62.82M	34.4	57.6	37.8	33.7	54.9	37.1	42.6	134.6G	30.9
BoxInst	Resnet-101	53.98M	38.0	62.1	41.4	26.8	50.3	23.8	40.4	115G	29.1
Discobox	Resnet-101	65.06M	33.3	56.3	34.0	28.2	50.5	28.5	38.5	124.7G	30.3
Apro_box2mask	Resnet-101	62.89M	44.0	66.7	48.1	26.6	53.9	22.7	43.7	78.5G	16.3
BoxSnake	Resnet-101	45.9M	52.1	79.5	59.2	42.8	67.9	46.5	58.0	136.6G	16.1
Box2Mask	Resnet-101	62.9M	48.8	74.0	52.7	33.5	60.5	32.6	50.4	78.9G	18.1
RSPrompter	ViT-L	320.2M	41.1	73.0	44.3	15.4	25.8	15.1	35.8	664.5G	20.8
BoxSnake	Swin-L	217.5M	58.1	**86.0**	67.4	44.5	68.0	49.1	62.2	607.1G	15.7
Box2Mask	Swin-L	215.5M	45.5	72.0	48.4	32.7	59.8	31.0	48.2	234.4G	14.3
Ours	ConvNeXt-V2-B	110.6M	**61.9**	84.9	**72.2**	**59.1**	**83.6**	**65.5**	**71.2**	172G	28.7

recognition under the 4% training set. While our method outperforms the baselines in vehicle detection, noticeable omissions still occur under the 7% training set.

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